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To cite this article before publication: Manodip Routh *et al* 2025 *J. Phys.: Condens. Matter* in press <https://doi.org/10.1088/1361-648X/ae0670>

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Exploring Quantum Phases in Frustrated Spin- $\frac{1}{2}$ Chain and Ladders: A Detailed Review

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Abstract. In this topical review, we explore low-dimensional spin- $\frac{1}{2}$ systems, focusing on key models such as the Majumdar-Ghosh (MG) model on a spin- $\frac{1}{2}$ chain and the Heisenberg spin- $\frac{1}{2}$ model applied to various ladder geometries, including normal, zigzag, two-leg honeycomb, skewed, and diamond. We discuss the key physical features and experimental realizations of these models in real materials, providing a detailed overview. The review also includes the Kitaev-Heisenberg model on a two-leg ladder, a recent development in quantum magnetism. It discusses emerging quantum phases, such as spin liquids and magnetization plateaus, and highlights open research areas, including the investigation of unexplored exotic ground states. Additionally, we briefly discuss how quantum simulators based on ion traps, superconducting circuits, optical lattices, etc., are being used as powerful tools to explore the ground-state phases of spin- $\frac{1}{2}$ systems. Furthermore, the review emphasizes the potential for new material design, offering an exciting avenue for future experimental and theoretical advancements in low-dimensional quantum systems.

Keywords: MG model, zigzag ladder, two-leg ladder, diamond ladder, cuprates, Kitaev-Heisenberg ladder

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1. Introduction

The development of microscopic spin models began with W. Lenz's efforts to provide a microscopic foundation for Weiss's molecular field theory, which describes cooperative behavior in magnetic systems [1, 2]. In 1925, E. Ising, a student of Lenz, investigated the one-dimensional (1D) version of this model, which consists of an infinite chain of spins interacting via nearest-neighbor coupling between their longitudinal components. This model, now famously known as the Ising model, has become a cornerstone of statistical mechanics [3, 4]. In contrast, the isotropic Heisenberg model, introduced by W. Heisenberg in 1928, represents the other extreme of spin interactions [5]. The ground states of 1D antiferromagnetic (AFM) uniform chains of $S = \frac{1}{2}$ spins exhibit remarkable differences between these two models. While the Ising model results in an ordered ground state, the Heisenberg model features a ground state with quasi-long-range order, even at $T = 0$ K.

The Onsager's famous solution of the Ising model extended to a two-dimensional (2D) square lattice, demonstrated the presence of long-range order at a finite temperature, with a transition temperature comparable to the exchange interaction J [6, 7]. In contrast, the 2D Heisenberg system remains disordered for $T \neq 0$ K but exhibits order at $T = 0$ K. The absence of long-range order in low-dimensional isotropic Heisenberg systems at finite temperatures can be understood, thanks to the Mermin-Wagner theorem [8], which asserts that true long-range order in the ground state of isotropic Heisenberg spin- $\frac{1}{2}$ systems is only possible in dimensions higher than two.

The ferromagnetic (FM) Heisenberg and Ising models have been widely explored across different dimensions, yielding important insights [9–11]. However, they are generally less appealing compared to their AFM counterparts. The antiferromagnetic versions of these models exhibit intriguing behavior, as their ground state is influenced by the lattice geometry. In certain lattices, such as square [12–15], triangular [16–20], kagome [21–29] and kagome-strips [30–34], competing exchange interactions among the spins make the system frustrated and leads to many exotic ground state.

The concept of frustration in magnetic systems was first introduced by G. Toulouse [35, 36]. These magnetic systems had been studied since the 1950s. Early works included G. H. Wannier's investigation of the Ising model on a triangular lattice with nearest-neighbor AFM exchange [37]. Frustrated magnetism is typically categorized into two types: *geometrical frustration* and *exchange frustration*. These types of frustration differ in their origins and can lead to distinct types of behavior, depending on the

dimensionality of the system.

In geometrically frustrated magnetic systems, the underlying lattice structure prevents the simultaneous minimization of all exchange interactions. A well-known example of this phenomenon occurs in a triangular lattice, where three spins form a fundamental frustrated unit [16–20]. Fig. 1(a) shows a triangular plaquette with three antiferromagnetic bonds, where if two Ising spins within the plaquette align antiferromagnetically to minimize their interaction energy, the third spin cannot align in a way that satisfies all pairwise interactions. However, in 1D and quasi-1D systems, geometrical frustration is typically less pronounced because the interaction is restricted to a linear or nearly-linear arrangement.

Exchange frustration arises from competing exchange interactions between spins, which prevent the system from simultaneously satisfying all the exchange interactions. Fig. 1(b) shows a square with three ferromagnetic bonds and one antiferromagnetic bond indicated by dashed and solid lines, respectively. Here also, one sees that if one tries to put Ising spins on those plaquettes, at least one of the bonds around the plaquette will not be satisfied. In this review, we have discussed about $J_1 - J_2$ model in 1D chain in Section 3, which is also a frustrated model where nearest-neighbor (NN) J_1 and AFM next-nearest-neighbor (NNN) J_2 interactions compete with each other as shown in Fig. 3(a).

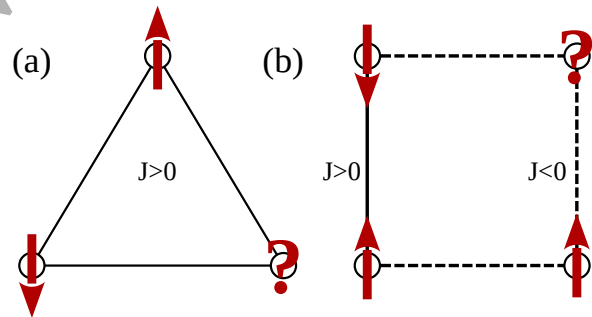


Figure 1. Examples of (a) geometrical and (b) exchange frustrated plaquettes. Ferromagnetic and antiferromagnetic couplings, denoted by $J < 0$ and $J > 0$, are represented by dashed and solid lines, respectively. Ising spins are indicated by $\uparrow$ and $\downarrow$. Regardless of the orientation chosen for the spin marked with a question mark, at least one of its bonds remains unsatisfied, indicating the presence of a frustrated bond.

For several decades, the investigation of low-dimensional frustrated magnetic systems remained predominantly theoretical due to the lack of experimentally realized low-dimensional materials, such as spin chains and ladders. This situation began to change in the 1970s with the synthesis of materials whose magnetic properties closely aligned with those predicted by abstract low-dimensional quantum spin models. The presence of strong quantum fluctuations in the low-

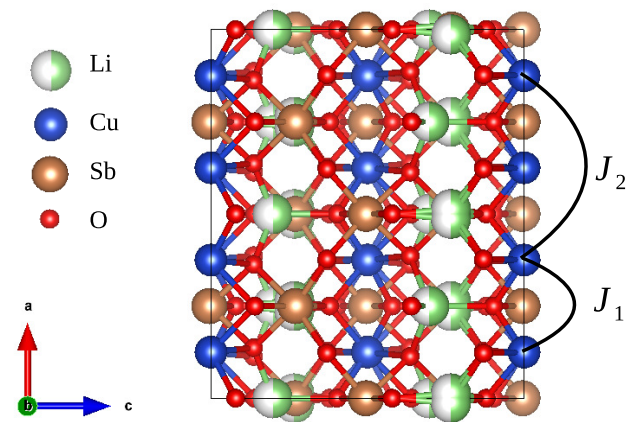


Figure 2. 3D crystallographic structure of LiCuSbO_4 . Edge-sharing CuO_4 octahedra form spin- $\frac{1}{2}$ chains along the a -axis, while Li atoms separate these chains along the c -axis. The nearest-neighbor (J_1) and next-nearest-neighbor (J_2) spin exchanges are depicted by black lines.

dimensional magnetic subsystems of high-temperature superconductors reignited significant interest in both theoretical and experimental investigations of quantum spin systems [38]. Subsequent studies on spin chains and ladders demonstrated that even purely insulating low-dimensional quantum magnets exhibit a wealth of intriguing phenomena.

One prominent candidate for the realization of low-dimensional quantum spin models is inorganic transition-metal (TM) compounds [39]. Materials are generally three dimensional in nature and realization of spin chains and ladders requires manipulation of location of magnetic atom in the crystal and controlling the exchange pathway of magnetic interactions. In many cases, materials are insulating and only the spin degree of freedom is active, effective magnetic exchanges decides the dimensionality of the system. In many TM compounds like cuprates and vanadates atoms are arranged closely along a particular direction relative to other directions and these leads to effective one dimensional magnetic system for example see Fig. 2, where the edge-shared CuO_4 plaquettes form 1D chain along a -axis where the AFM nearest neighbor and next-nearest neighbor exchange interactions are J_1 and J_2 , respectively. The Cu-atoms in other directions are well separated compared to atoms along the a -axis.

Low-dimensional frustrated magnetic systems have gained significant attention due to advances in synthesis techniques for low-dimensional materials, as well as the theoretical predictions of exotic quantum phases and novel excitations in these systems [40–42]. These novel phases and phenomena arise due to the interplay of quantum fluctuations in confined nature of the geometry and competing exchange interactions. The enhanced quantum fluctuations in these system

give rise to many exotic quantum phases like quantum spin liquid [43–46], vector chiral [47–49], Néel [50, 51], dimer order [47], topological ground state [52–54] etc. These effects can also lead to emergence of the novel excitations like fractionalized spinons [55, 56], psinons [57] and magnons [58, 59], Majorana fermion [54, 60–64] in one dimensional spin- $\frac{1}{2}$ system, whereas, doublon and quarton in trimer [65, 66] etc. These systems can also have many interesting thermal properties.

The study of low-dimensional frustrated magnets continues to be an active area of research, with several reviews summarizing progress in the field. For instance, Dagotto et al. have reviewed aspects of ladder and two-dimensional systems in Refs. [67, 68]. However, to the best of our knowledge, a detailed review focusing specifically on theoretical and experimental studies of 1D and quasi-1D spin- $\frac{1}{2}$ magnetic systems remains lacking in the literature.

In the topical review, we discuss the frustrated spin- $\frac{1}{2}$ model system on chain and some of the selected two leg ladder geometries, especially, those models which are already experimentally realized and some model systems which show exotic quantum phases and yet to be realized but are of current interest. We also provide table of materials corresponding to relevant model and their parameters.

We have tried our best to provide the list of materials which are generally well accepted as realization of model systems. Isotropic Heisenberg spin- $\frac{1}{2}$ systems in 1D chain, and on the zigzag, normal, honeycomb, diamond lattice ladder systems are the main focus of our review. We also review two new recent model systems: Kitaev-Heisenberg model on two leg ladder and isotropic AFM model on skewed ladder geometries. Both of these systems are new and interest of current research and however, experimental realization is still awaited.

Furthermore, a brief review of how spin- $\frac{1}{2}$ systems are realized on quantum simulators is provided. In particular, we highlight how quantum simulators based on ion trapping, superconducting circuits, optical lattices etc. are used as powerful tools to explore the ground state phases like topologically ordered phase, spin ice, spin glasses, quantum spin liquids and quantum scars. We also add a brief introduction of some well known quantum information probes such as bipartite entanglement entropy, fidelity, and concurrence are discussed in the context of diagnosing quantum phase transitions.

Plan of the review: This review is organized into eleven sections. Section 2 discusses about the theoretical tools used for quantum spin systems. Next we focus on specific type of ladder geometry. Section 3 begins with an examination of the frustrated isotropic Heisenberg spin- $\frac{1}{2}$ chain, which incorporates

both nearest-neighbor and next-nearest-neighbor interactions. This system can also be interpreted as a zigzag ladder. Each section introduces the model on the corresponding lattice, followed by a detailed analysis of its quantum phase diagram and experimental realizations. Sections 4, 5, 6, 7, and 8 discuss the quantum phases of models on a standard two-leg ladder, a two-leg honeycomb ladder, a two-leg ladder with weak inter-ladder coupling, a diamond ladder, and a skewed ladder, respectively. Reflecting recent advancements, Section 9 explores the Kitaev-Heisenberg model on a two-leg ladder and its exotic quantum phases. Finally, Section 10 presents an outlook on potential future directions in this field.

2. Theoretical tools for quantum spin systems

In this review we will discuss about various exotic phases in frustrated spin systems. Many of these phases can be described using quantum many-body models, where quantum interactions play a crucial role in determining the underlying physics. However, only a few quantum many-body systems can be solved analytically. For instance, in the case of the one-dimensional Heisenberg model, certain low-energy states can be determined exactly using the Bethe ansatz [69–71] and references therein.

Solving a quantum many-body Hamiltonian is inherently challenging due to the exponential growth of the Hilbert space with system size. Specifically, for a spin- $\frac{1}{2}$ system with N sites, the dimension of the Hilbert space scales as 2^N . Given this rapid increase in computational complexity, exact solutions are often impractical for large systems.

To address this challenge, approximate methods, both analytical and numerical, are widely employed. Analytical techniques include spin-wave theory [72–78] and references therein, field-theoretical approaches such as bosonization [79–82] and references therein, and the renormalization group framework [83–88] and references therein. These methods have been extensively used in the study of quantum magnetism, particularly in low-dimensional systems. However, their applicability is generally restricted to specific models, and for more complex systems, they often fail to provide reliable results.

In such cases, numerical methods serve as powerful alternatives. These methods can be broadly classified into two categories: exact numerical techniques applicable to finite systems and approximate numerical approaches. The exact diagonalization (ED) [89, 90] numerical technique allows us to solve the many body models with small system size, i.e., with small Hilbert space dimension. For large system sizes several approximate numerical techniques have been

developed, i.e., quantum monte carlo (QMC) [89–96] and references therein, numerical renormalization group (RG) [90, 97, 98] and references therein, density matrix renormalization group (DMRG) [99, 100], tensor network [101], which have proven to be effective in exploring the physics of strongly correlated quantum systems. Among all the techniques, DMRG algorithm has gained widespread popularity in condensed matter physics due to its ability to handle strongly correlated systems. It has been applied to study the ground-state properties of 1D and quasi-1D magnets, as well as fermionic and bosonic systems. References [101–110] and references therein provide detailed reviews of the DMRG method and its applications. Tensor network is also very powerful tool to solve many-body physics and detailed reviews of this method are provided in References [109, 111–113] and references therein.

3. Frustrated spin- $\frac{1}{2}$ chain and zigzag ladder

The 1D FM isotropic Heisenberg spin- $\frac{1}{2}$ chain was initially studied by W. Heisenberg in 1928 [5]. Subsequently, in 1931, H. Bethe exactly solved the ground state of the 1D AFM isotropic spin- $\frac{1}{2}$ Heisenberg model [69], revealing a gapless continuum of spinon with the algebraic decay of spin-spin correlation [114, 115].

In 1969 Chanchal K. Majumdar and Dipan K. Ghosh introduced a model with nearest and next-nearest neighbors [116, 117]. The model Hamiltonian can be written as:

$$\mathcal{H}(J, \alpha) = \frac{J}{2} \sum_i \vec{S}_i \cdot \vec{S}_{i+1} + \frac{J\alpha}{2} \sum_i \vec{S}_i \cdot \vec{S}_{i+2}, \quad (1)$$

where, the nearest neighbor exchange interaction $J/2$ set the energy scale and α represents the ratio of the nearest and next-nearest neighbor exchange and $\vec{S}_i$ are spin- $\frac{1}{2}$ vector at site i . For $\alpha = 1/2$ they proposed that the system has two degenerate ground states, with spins arranged in nearest-neighbor singlet pairs. These two states are connected by a inversion symmetry. However the model now rewritten as [118]:

$$\mathcal{H}(J_1, J_2) = J_1 \sum_i \vec{S}_i \cdot \vec{S}_{i+1} + J_2 \sum_i \vec{S}_i \cdot \vec{S}_{i+2}, \quad (2)$$

where first and second term refer nearest and next-nearest neighbor spin exchange interactions. This model exhibits frustration for AFM J_2 ($J_2 > 0$) and either sign of J_1 . The frustration parameter, defined as $\alpha = J_2/J_1$, quantifies the competition between the first and second neighbor couplings. The system has one spin per unit cell and the total spin $S = \sum_i S_i^z$ is conserved along with its z -component. The arrangement of spin is shown in Fig. 3 (a) and in large

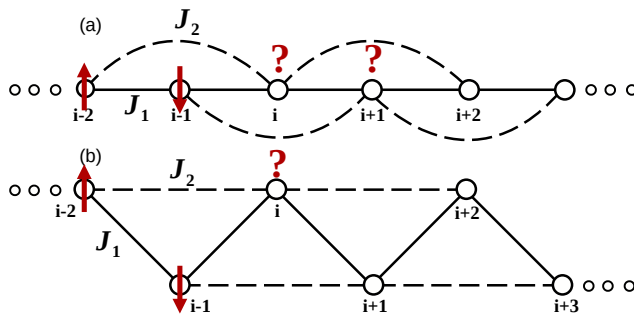


Figure 3. (a) Schematic representation of the one-dimensional chain with nearest-neighbor (J_1) and next-nearest-neighbor (J_2) interactions. (b) In the large J_2 limit, the chain can be viewed as a zigzag ladder, with odd and even sites forming the upper and lower legs, respectively. In both cases, red arrows are indicating the magnetic frustration in the system.

J_2/J_1 limit this structure can be mapped to a zigzag chain [119].

Quantum phases: The ground-state quantum phase diagram of Eq. (2) in the J_1 – J_2 parameter space is shown in Fig. 4, revealing six primary quantum phases: two types of spin liquid phases—one with quasi-long-range order (QLRO) at ($q = \pi$) and another with decoupled gapless quasi-long-range order at ($q = \pi/2$); a gapped dimer phase; gapped spiral-I and spiral-II phases; and a ferromagnetic (FM) phase.

By tuning J_2/J_1 we can access all these phases and phase boundaries can be calculated using quantities such as the energy derivative, spin correlation function and energy level crossover [120]. The Lieb-Schultz-Mattis (LSM) theorem states that for an

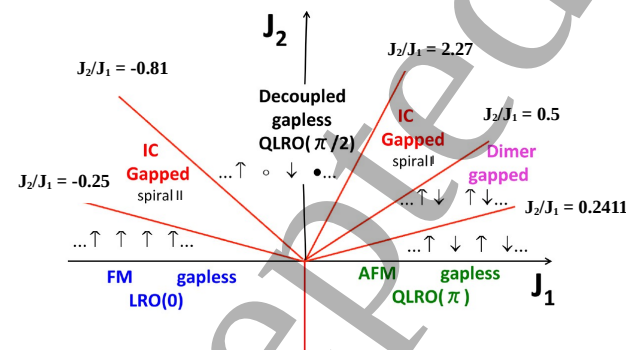


Figure 4. The quantum phase diagram of the Hamiltonian $H(J_1, J_2)$, as depicted in Eq. (2), displays the critical points for the ratio J_2/J_1 at -0.25 , -0.81 , 2.27 , and 0.2411 . The exact location of $J_2/J_1 = -0.25$ separates a gapless ferromagnetic (FM) phase from a gapped incommensurate (IC) phase. Between $J_2/J_1 = -0.81$ and $J_2/J_1 = 2.27$, a gapless decoupled phase is observed, where open and closed circles represent spins oriented in and out of the plane, respectively. The gapped IC phase continues until the MG point, $J_1 = 2J_2$, and the dimer phase persists until $J_2/J_1 = 0.2411$, beyond which the system transitions into a gapless antiferromagnetic (AFM) phase. This Fig. is reproduced and modified from the ref. [118].

isotropic Heisenberg half-integer spin system, in the thermodynamic limit, either the lowest excitation is gapless or ground state must be degenerate [121, 122]. However, in a finite-size system, the lowest excited state can have a finite gap and it can be either singlet or triplet. In thermodynamic limit, either the triplet excitation gap vanishes and the spectrum becomes gapless, or the singlet excited state becomes degenerate with the ground state. This finite gap or quasi-degeneracy arises due to finite-size effect. Let us now first discuss these phases in detail.

Quasi-long range order (QLRO): At $J_2 = 0$, $J_1 > 0$ limit, the model in Eq. (2) reduced to a Heisenberg antiferromagnetic (HAF) spin- $\frac{1}{2}$ chain. The ground state exhibits an antiferromagnetic spin arrangement, as illustrated in Fig. 3 (a). The energy spectrum is gapless [123]. Luther and Peschel [124] first predicted the critical exponents for spin-spin power-law correlations in 1975, which agreed with Bonner and Fisher’s seminal numerical results from 1964 [125]. In 1989, a multiplicative logarithmic correction to the power-law decay was introduced due to a marginally irrelevant “spin-Umklapp” process [126–129], later verified numerically [119, 120, 130–141]. The spin correlation function is given by

$$\langle S(r) \cdot S(0) \rangle \propto \frac{(-1)^r \log(r)^{1/2}}{r}. \quad (3)$$

This phase is referred to as a spin liquid, also known as QLRO ($q = \pi$), which signifies quasi-long-range order in the spin correlations at the wave vector $q = \pi$. Sandvik has reviewed numerical studies of the HAF and related spin chains, at $\alpha < 1$ [89]. Lecheminant addressed frustrated 1D spin system mainly from field theory perspective for $\alpha < 1$ [142]. The spectrum of this model in small α limit has been extensively studied and lowest excitation is triplet state was predicted in the early studies of this model [143–145].

In another extreme limit, where $J_1 \sim 0$ and $J_2 > 0$, the system can be viewed as a zigzag ladder, with all odd and all even sites forming the two legs of this ladder, connected by J_2 exchanges. The J_1 exchanges serve as the rungs between the legs, as illustrated in Fig. 3 (b). In absence of rung the ladder acts as two decoupled chains, with each chain behaving as a HAF spin chain exhibiting an antiferromagnetic spin configuration. The spin correlation follows the power law behavior with wave vector at $q = \pi/2$. This phase is referred to as QLRO ($\pi/2$). It persists for $-0.81 < J_2/J_1 < 2.27$ [118, 146].

Dimer phase: The doubly degenerate ground state and its wave function was predicted by C. K. Majumdar and D. Ghosh at $J_2/J_1 = 1/2$ and it is

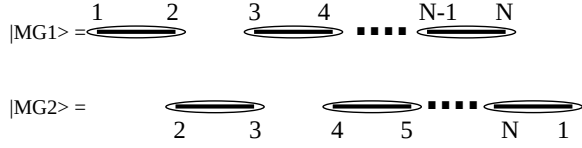


Figure 5. Doubly degenerate dimerized ground states ($|\text{MG1}\rangle$ and $|\text{MG2}\rangle$) of the MG model are shown and solid bonds indicate the dimer singlet of two neighboring spins.

written by

$$|\Psi_1^{\text{MG}}\rangle = \prod_{i=1}^{N/2} [2i-1, 2i]$$

$$|\Psi_2^{\text{MG}}\rangle = \prod_{i=1}^{N/2} [2i, 2i+1], \quad (4)$$

where $[i, j]$ singlet dimer $[i, j] = \frac{1}{\sqrt{2}}(\uparrow_i \downarrow_j - \downarrow_i \uparrow_j)$. Pictorial representation of the ground states are shown in Fig. 5. The ground state energy is $E^{\text{MG}} = -\frac{3J_1 N}{8}$, and the singlet-triplet (ST) energy gap is exactly $\Delta = J_1$ [147, 148]. The gapless spin liquid to gapped dimer transition was first predicted in 1983 by F. D. M. Haldane at $J_2/J_1 = 0.3$ [149]. Later, it was shown numerically that the critical point is at $J_2/J_1 = 0.2411$ by Okamoto and Nomura using the level crossing method [120]. DMRG calculations also supported this claim [119, 136, 139].

White and Affleck predicted the ST gap using a combination of DMRG and field theory, which is given by [136]:

$$\Delta = \frac{v_s}{\xi} \propto \exp(J_1/(J_2 - J_{2c})), \quad (5)$$

where v_s , ξ are the velocity and correlation length in the system, respectively and $J_{2c} = 0.2411$. White et al. concluded that this phase extends to $J_1 = 0$. Itoi and Qin presented a more detailed field theory for large J_2/J_1 [138] and showed that the spin correlation length diverges as:

$$\xi(J_1, J_2) \sim \exp \left[c \left(\frac{|J_1|}{J_2} \right)^{-2/3} \right] \quad (6)$$

The spin correlation function and ST gap in this phase are extensively studied by various groups [119, 136, 139, 140]. This phase is also characterized by spontaneous dimerization in the system, which can be expressed as:

$$B = \frac{1}{N} \sum_n (-1)^n \langle S_n \cdot S_{n+1} \rangle. \quad (7)$$

It is the average difference between the nearest-neighbor correlations. The ST gap Δ and the

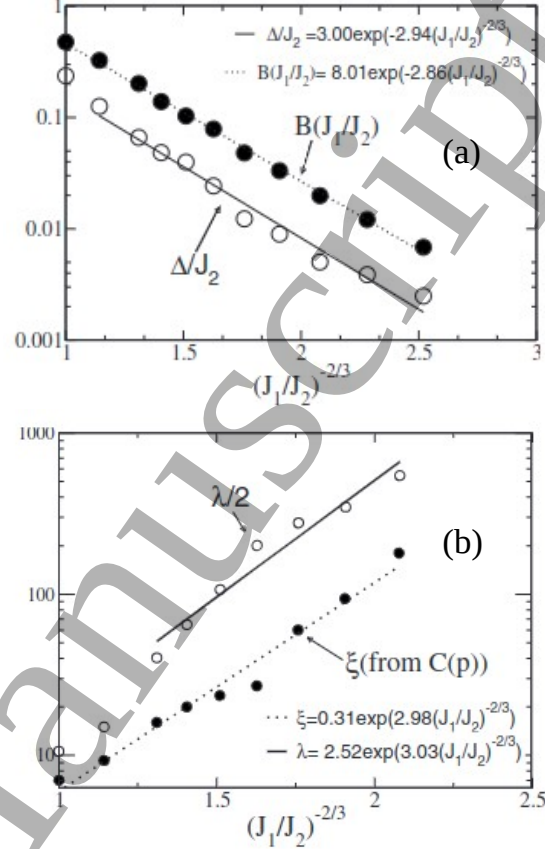


Figure 6. (a) Order parameter B and singlet-triplet energy gap Δ as a function of $(\frac{J_1}{J_2})^{-2/3}$. The fitted lines are $1/\epsilon$ in Eq. (6) with $c = 2.94$ for Δ and 2.86 for B . (b) The spin correlation length, ξ , of the bond order wave (BOW) phase and the wavelength, λ , of the spiral phase are functions of $(\frac{J_1}{J_2})^{-2/3}$. This figure is in large J_2/J_1 limit ($1 < J_2/J_1 < 4$). This Fig. is reproduced from Ref. [140].

dimerization order parameter B in the system are shown in Fig. 6(a), and the correlation length is shown in Fig. 6(b) [139, 140]. The ST gap Δ increases rapidly with J_2/J_1 and attains a maximum value at $J_2/J_1 \sim 0.55$, after which it decreases exponentially, as shown in Fig. 5 of Ref. [150]. The bond order follows a similar pattern, as shown in Fig. 6 of Ref. [139]. The dimer phase extends from $J_2/J_1 = 0.2411$ to 2.27 , as shown in Fig. 4.

Around the MG point, the system continues to exhibit spontaneous dimerization; however, the two singlet states with momenta $q = 0$ and $q = \pi$ are no longer exact eigenstates. In a finite chain with periodic boundary conditions, these two states are closely spaced in energy and are referred to as quasi-degenerate states. The energy gap between these singlet states decreases with increasing system size N , and in the thermodynamic limit ($N \rightarrow \infty$), this gap

vanishes. This leads to spontaneous dimerization and twofold ground-state degeneracy [139].

Gapped spiral I & II: In the AFM regime ($J_1 > 0$), the gapped spiral I phase emerges at $J_2/J_1 \approx 0.55$ and persists until the decoupled phase at $J_2/J_1 = 2.27$. A similar gapped spiral phase, referred to as gapped spiral II, exists in the FM regime ($J_1 < 0$) and spans the range $J_2/J_1 = -0.25$ to $J_2/J_1 = -0.81$. The ground state of both phases is dimerized, characterized by a finite ST gap $\Delta > 0$ and exponentially decaying correlations [118, 138]. However, the ST gap Δ and dimerization in the ferromagnetic regime are significantly smaller compared to their antiferromagnetic counterparts. Within these phases, the nearest-neighbor spins align with a specific pitch angle or spin wave vector q_G , determined by the J_2/J_1 ratio [119, 135, 139, 140, 150, 151].

The competition between AFM nearest-neighbor and next-nearest-neighbor exchange interactions causes the twisting of spin orientations. This behavior can be understood using the classical spin model described in Eq. (2), where the Hamiltonian energy is expressed as $E = J_1 \cos(\theta) + J_2 \cos(2\theta)$. Minimizing this energy yields the pitch angle, which can be shown to satisfy $\cos \theta = -J_1/4J_2$ [136].

In the classical system, θ changes continuously with J_2/J_1 , whereas in the quantum model, it changes only when $J_2/J_1 \approx 0.55$, and the transition is much sharper compared to the classical case. The pitch angle can be calculated using the structure factor [118] or directly from the correlation function [150].

DMRG and static structure factor $S(q)$ peaks yield q_G in large systems. The evolution of spin wave vector q_G with J_1/J_2 in gapped spiral phases is shown in Fig. 7 for system sizes of $N = 48, 96$ and 144 . The point $J_1 = 0$ is exact, as are $q_G = 0$ and π for $J_1/J_2 \leq -4$ and ≥ 2 , respectively. The plateau is reached at $J_1/J_2 = -1.24$ for $J_1 < 0$ and is left at $J_1/J_2 = 0.44$ for $J_1 > 0$. The points are extrapolated in the insets of Fig. 7.

To conclude the analysis of q_G in the spiral phases, we derive the solid line shown in Fig. 7. Classical spins account for the square root behavior at the exact point $J_1/J_2 = -4$. Based on general field-theoretical arguments, Nomura and Murashima [152] proposed that the wave vector $q \propto (J_2/J_1 - 1/2)^{1/2}$ near the commensurate-incommensurate transition, with $J_2/J_1 \geq 1/2$. Instead, we expand around small J_1/J_2 , noting that the ground-state correlations from Eq. (2) yield $q_G = 0, \pi/2$, and π for $J_1/J_2 \leq -4, 0$, and ≥ 2 , respectively. The combination of a square-root singularity at these exact points and the $J_1 = 0$ constraint, where $q_G = \pi/2$, leads to the expressions.

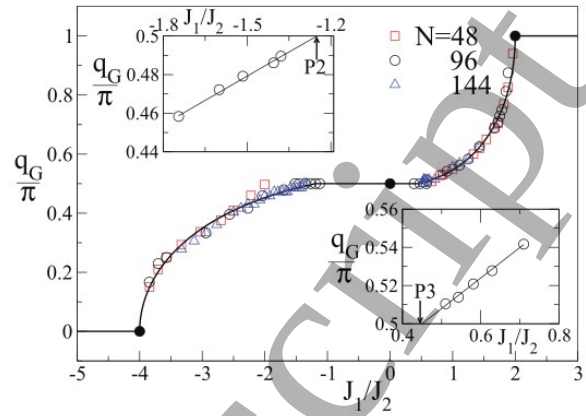


Figure 7. DMRG results for the wave vector q_G of ground state correlations as a function of J_1/J_2 in models of N spins in Eq. (2). Closed circles are exact in the thermodynamic limit. The insets show the J_1/J_2 values at which the $q_G/\pi = \frac{1}{2}$ plateau is reached and left in a step of $\frac{2}{N}$ for $J_1 < 0$ and $J_1 > 0$; linear extrapolation gives the quantum critical points P2 ($J_1/J_2 = -1.24$) and P3 ($J_1/J_2 = 0.44$). The intervals $(-4 \leq J_1/J_2 \leq -1.24)$ and $(0.44 \leq J_1/J_2 \leq 2)$ with variable q_G are spiral gapped phases with degenerate ground state. The commensurate phase between -1.24 and 0.44 has a non degenerate ground state. The solid black line in main figure is represented in Eq. (8). The Fig. is reproduced from Ref. [118].

$$\begin{aligned} \frac{q}{\pi} &= A \left(4 + \frac{J_1}{J_2} \right)^{1/2} \exp \left(-a \left(4 + \frac{J_1}{J_2} \right) \right) \leq \frac{1}{2}; \\ &\quad -4 \leq \frac{J_1}{J_2} \leq -1.24 \\ \frac{q}{\pi} &= \frac{1}{2}; \quad -1.24 \leq \frac{J_1}{J_2} \leq 0.44 \\ \frac{q}{\pi} &= 1 - B \left(2 - \frac{J_1}{J_2} \right)^{1/2} \exp \left(-b \left(2 - \frac{J_1}{J_2} \right) \right) \geq \frac{1}{2}; \\ &\quad 0.44 \leq \frac{J_1}{J_2} \leq 2 \quad (8) \end{aligned}$$

The corresponding curve from $q_G = 0$ to $-\pi (= \pi)$ is q_G/π . The parameters in Fig. 7 are $A = 0.38$, $B = 0.565$, $a = 0.086$ and $b = 0.22$. A and B are refer to singularities that are known in the thermodynamic limit; a and b are negligible near the exact points but they matter for critical points -1.24 and 0.44 that delimit the $q_G = \pi/2$ phase. Eq. (8) combines singularities at exact points and with the translational symmetry of the $J_1 - J_2$ model before making a continuum approximation.

Ferromagnetic phase: This is a trivial phase where all the spins are aligned in the same direction. However, as we increase the antiferromagnetic J_2 , the frustration in the system grows, leading to a first-order transition to a singlet state at $J_2/J_1 = 1/4$. Badar et al. demonstrated that under the ferromagnetic

condition, the ratio satisfies $|J_1|/J_2 > \cos^2(\frac{\pi}{N})$, where N denotes the size of the one-dimensional chain [153]. Later, the transition from the ferromagnetic to the incommensurate phase at $J_2/J_1 = -0.25$ was studied by various groups [154–157].

Experimental realization of the 1D chain and zigzag ladder: Spin chain systems provide a vital connection between advanced concepts in many-body quantum physics and experimentally measurable properties in real materials. Furthermore, we demonstrate that frustrated magnets, characterized by a macroscopic number of quasi-degenerate states in competition, serve as an exceptional platform for exploring emergent exotic phenomena [41]. Notably, the interplay between geometric frustration and confined dimensionality facilitates the emergence of unexpected quantum phases, often associated with the spontaneous breaking of continuous or discrete symmetries, a phenomenon referred to as “order by disorder”.

In this section, we focus on the realization of the MG model in real materials. As noted earlier, low-dimensional spin- $\frac{1}{2}$ structures are commonly found in three-dimensional transition metal (TM) oxides. These materials are typically insulating in nature and exhibit localized magnetic moments. For instance, a material containing copper in the +2 oxidation state (Cu^{2+}) effectively exhibits spin- $\frac{1}{2}$. The magnetic interaction between two neighboring Cu spin- $\frac{1}{2}$ moments can arise either from direct overlap of the d -orbitals, resulting in direct exchange, or through nonmagnetic ligands, leading to an indirect exchange process. Such ligands can include individual atoms such as oxygen, sulfur, or halides, or more complex molecular systems [39].

Copper oxide-based structures can be broadly classified into two categories based on the connectivity of their CuO_6 octahedra. In corner-sharing CuO_6 octahedra, each octahedron shares only a corner (vertex) with its neighbors. This arrangement facilitates significant overlap between the p -orbitals of oxygen and the $3d$ orbitals of adjacent Cu ions, resulting in stronger electronic interactions. Consequently, these materials exhibit stronger magnetic exchange interactions. Notable examples include Sr_2CuO_3 and SrCuO_2 [158].

In contrast, edge-sharing CuO_6 octahedra share one edge with neighboring octahedra, leading to a bonding environment where the orbitals of adjacent Cu ions are nearly orthogonal. This configuration limits orbital overlap and weakens direct electronic interactions, thereby reducing the dimensionality of magnetic interactions and weakening the exchange coupling.

The Goodenough-Kanamori-Anderson (GKA) rule [159–162] suggests that the bond angle is also crucial in determining the nature of the exchange interaction.

When the Cu-O-Cu bond angle is close to approximately 90° , it favors the superexchange mechanism, resulting in a ferromagnetic interaction. In contrast, when the bond angle is close to 180° , it leads to an antiferromagnetic exchange.

Several other transition metal (TM) atoms may also exhibit an effective spin- $\frac{1}{2}$, depending on factors such as the ionic state of the atom, crystal field splitting, and other environmental conditions. For instance, vanadium-based compounds [163–172] can form atomic doublets or effective spin- $\frac{1}{2}$ states, exhibiting behavior analogous to the spin- $\frac{1}{2}$ state of Cu^{2+} . In many cuprates, the next-nearest-neighbor spin exchange, J_2 , is finite due to possible exchange pathways such as Cu-O-O-Cu, which typically result in antiferromagnetic interactions. As previously discussed, a finite antiferromagnetic J_2 introduces frustration irrespective of the nature of the nearest-neighbor exchange, J_1 . The magnitude of J_2 is influenced by the effective orbital overlaps between next-nearest-neighbor atoms.

The exchange interactions in these materials can be quantitatively determined by fitting experimental data, such as magnon dispersion obtained from neutron scattering measurements, specific heat, and magnetic susceptibility. This section focuses on spin- $\frac{1}{2}$ quasi-one-dimensional chain or zigzag ladder materials and their modeling using the MG model, emphasizing their J_1 and J_2 parameters.

There are numerous materials that exhibit antiferromagnetic J_1 and J_2 interactions, such as Sr_2CuO_3 [158,173–175] and SrCuO_2 [158,174,176–178]. The Table 1 provides the list of materials which have the antiferromagnetic J_1 and J_2 .

In ferromagnetic J_1 of the $J_1 - J_2$ model spin- $\frac{1}{2}$ chain are generally realized in edge-shared cupric octahedron CuO_6 and the bond angle Cu-O-Cu is generally 90° . $J_2/|J_1|$ in these system may ranges from [186, 187] $\alpha \approx 0$ in $\text{Ba}_3\text{Cu}_3\text{In}_4\text{O}_{12}$ or $\text{Ba}_3\text{Cu}_3\text{Sc}_4\text{O}_{12}$ to $\alpha \approx 0.5$ in $(\text{N}_2\text{H}_5)\text{CuCl}_3$, [184] LiCu_2O_2 , [188] LiCuSbO_4 , [189] LiCuVO_4 , [190] and $\text{Rb}_2\text{Cu}_2\text{Mo}_3\text{O}_{12}$ [191] and LiCuVO_4 [192].

LiCuVO_4 is a well-studied system exhibiting ferroelectricity at low temperatures and spin nematicity under high magnetic fields. The crystal structure of this material is illustrated in Fig. 8(b). The Cu^{2+} ions form magnetic chains with spin $S = \frac{1}{2}$ along the orthorhombic b axis [192]. The coupling constants are determined to be $J_1 = -19$ K and $J_2 = 44$ K. However, to accurately describe the long-range helical order that emerges at the Néel temperature $T_N = 2.4$ K, it is necessary to incorporate an interchain coupling $J_0 = -4.6$ K [193]. The incommensurate wavevector is identified as $Q = (0, 0.532, 0)$, and the spin spiral ground state lies within the ab plane.

Table 1. Table for materials related to zigzag ladder with AFM nearest neighbor J_1 and AFM next nearest neighbor J_2 .

Materials	References	Model parameters (Kelvin)	Experimental observation
CuGeO ₃	[179–182]	$J_1 = 150 - 160$, $J_2 = 36 - 57.6$; $J_1 = 160$, $J_2 = 56$, $\delta = 0.17^a$	Susceptibility, specific heat [179, 180]; Inelastic neutron scattering (INS) [181]
Cu(ampy)Br ₂	[183]	$J_1 = 17$, $J_2 = 3.4$	Susceptibility and high-field magnetization [183]
(N ₂ H ₅)CuCl ₃	[184]	$J_1 = 4.1$, $J_2 = 16.3$	Susceptibility, specific heat, Electron spin resonance (ESR) [184]
Cu ₆ Ge ₆ O ₁₈ - 6H ₂ O	[185]	$J_1 = 222$, $J_2 = 60$	Susceptibility and magnetization [185]

^a δ = dimerization. See the text (Spin-Peierls materials section) for details.

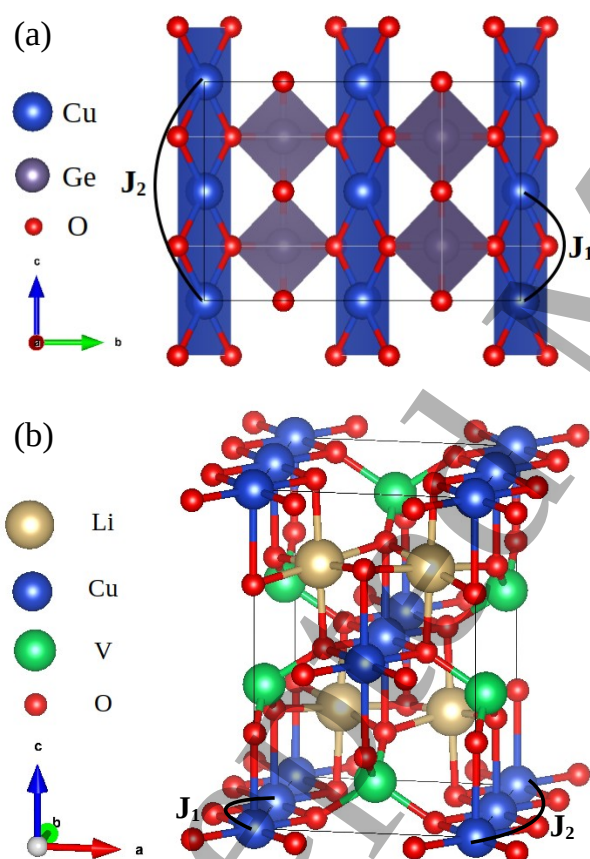


Figure 8. 3D crystallographic structure of (a) CuGeO₃ and (b) LiCuVO₄. Edge-sharing CuO₄ octahedra form spin- $\frac{1}{2}$ chains along the c -axis in CuGeO₃ and along the b -axis in LiCuVO₄. The nearest- and next-nearest-neighbor spin exchanges, J_1 and J_2 , are represented by black lines. The nearest-neighbor spin exchange J_1 is antiferromagnetic in CuGeO₃ and ferromagnetic in LiCuVO₄.

LiCuSbO₄ is also a well-studied spin- $\frac{1}{2}$ chain material that exhibits ferromagnetic nearest-neighbor interaction J_1 and a frustrated ratio of ferromagnetic to antiferromagnetic interactions, $J_2/J_1 = -0.45$ [189]. This system does not exhibit three-dimensional ordering down to 100 mK. It also displays a spin nematic phase under the influence of an external magnetic field, along with weak anisotropy in the exchange interactions [194]. In contrast to LiCuVO₄, which demonstrates long-range order at low temperatures, LiCuSbO₄ does not show any such ordering down to $T_N = 0.1$ K.

Another spin- $\frac{1}{2}$ chain compound exhibiting a diverse range of magnetic phases is the linarite compound PbCuSO₄(OH)₂. Linear spin wave theory predicts the exchange couplings to be $J_1 = -114$ K and $J_2 = 37$ K [195], although experimental observations indicate a weak interchain coupling of $J_0 = 4$ K. DMRG calculations suggest an anisotropic behavior for the exchange interaction, with $J_1 = (J_{1x}, J_{1y}, J_{1z}) = (-91.1, -86.6, -88.4)$ K and $J_2 = 28.3$ K, providing a more accurate description of the experimental results. These calculations also predict an interchain coupling of $J_0 = 2.7$ K [196]. Several studies have been conducted to investigate the magnetic properties of this material [197–199].

The compound A₂Cu₂Mo₃O₁₂ (A = Rb or Cs) represents another one-dimensional spin- $\frac{1}{2}$ Heisenberg system. For A = Rb, the system exhibits a ferromagnetic exchange interaction with $J_1 = -138$ K and an antiferromagnetic interaction with $J_2 = 51$ K, while for A = Cs, the values of J_1 and J_2 are -93 K and 33 K, respectively [200, 201]. Rb₂Mo₃Cu₂O₁₂ serves as an ideal candidate for investigating materials near the critical value of $J_2/J_1 = 0.25$ [191, 200, 202, 203]. The ratio J_2/J_1 suggests that Rb₂Mo₃Cu₂O₁₂ possesses a spin-singlet incommensurate (IC) ground state. H.

T. Lu et al. [204] computed the thermodynamic properties, such as susceptibility and specific heat, as well as the field dependence of the magnetization using the transfer-matrix renormalization group method.

A list of materials exhibiting ferromagnetic J_1 and antiferromagnetic J_2 interactions is provided in Table 2. This study primarily focuses on Cu-based one-dimensional and quasi-one-dimensional materials.

Dimerized $J_1 - J_2$ model: In certain cases, the interaction between electrons and lattice vibrations can result in lattice dimerization at low temperatures. Assuming that the dimerization affects only the nearest-neighbor exchange interactions, the modified form of the dimerized MG model can be expressed as follows:

$$\mathcal{H} = \sum_n J_1(1 + (-1)^i \delta)(\vec{S}_i \cdot \vec{S}_{i+1}) + J_2(\vec{S}_i \cdot \vec{S}_{i+2}) \quad (9)$$

The parameter δ represents the dimerization in the nearest-neighbor exchange interactions. The ground state of the dimerized antiferromagnetic Heisenberg spin- $\frac{1}{2}$ chain is a non-degenerate singlet, and the Hamiltonian exhibits a gap between the singlet ground state and the lowest triplet state for any finite value of δ . This model was first studied by Bray et al. in 1975 to describe the magnetic properties of a specific material within the mean-field approximation [237]. Subsequent theoretical investigations by various groups have provided significant insights into its properties [238–241]. Cross and Fisher analyzed the dimerized Heisenberg spin- $\frac{1}{2}$ chain using the mean-field random-phase approximation [242, 243]. The scaling theory predicts that the spin dimerization energy $\epsilon(\delta)$ and the gap $\Delta(\delta)$ scale as $\delta^{4/3}$ and $\delta^{2/3}$, respectively [242, 243]. However, experimental results have shown deviations from these scaling predictions for the energy gap [244–248].

The modeling of dimerized antiferromagnetic spin- $\frac{1}{2}$ Heisenberg chains remains a challenging problem. Recently, hybrid methods combining exact diagonalization and DMRG have emerged as accurate tools for investigating the magnetic properties of spin-Peierls and dimerized spin systems [249, 250].

Spin-Peierls materials: The spin-Peierls (SP) transition in the antiferromagnetic spin- $\frac{1}{2}$ system is analogous to the Peierls transition in fermionic systems [251–253]. This transition is characterized by a lattice dimerization at low temperatures, which arises due to spin-lattice couplings. In the adiabatic limit, where phonons follow spins without relaxation, the conventional approach is not sufficient for describing the SP transition. The flow equation method has been employed to reduce the general spin-phonon model to a spin-only Hamiltonian [254, 255], with the spin-Peierls gap emerging from the combined effects of alternation and frustration. The inclusion

of explicit spin-phonon coupling introduces several features that extend beyond the approximation of explicitly dimerized spin chains [256].

In fermionic systems, dimerization leads to the opening of a gap at the Brillouin zone boundary in the electronic spectrum. A similar phenomenon occurs in spin chains, where dimerization stabilizes the ground state energy and opens an energy gap [257, 258].

The spin-Peierls transition was initially discovered in the organic compound TTF-CuS₄C₄(CF₃)₄ at $T_{SP} = 12$ K [237, 259]. The spin- $\frac{1}{2}$ chain at $T > T_{SP}$ has equally spaced TTF⁺ ion and is dimerized below $T < T_{SP}$. This compound is modeled by linear Heisenberg antiferromagnetic chain with equal exchange J_1 to both neighbors for $T > T_{SP}$ and alternating exchange $J_1(1 \pm \delta(T))$ for lower T . Subsequently, Hase et al. found that the reduction in magnetic susceptibility, $\chi(T)$ for an inorganic compound CuGeO₃ [260] with $T_{SP} = 14$ K. Its crystal structure is depicted in Fig. 8(a). By analyzing magnetic susceptibility for $T > T_{SP}$, Riera and Dobry estimated $J_2/J_1 = 0.35$ with $J_1 = 160$ K [261]. However, the extensive studies of $\delta(T)$ for this compound remains inconclusive [262–269]. Small interchain couplings were also detected using inelastic neutron scattering, with $J_b = 0.1J$ and $J_c = -0.1J$ [181]. Sudip et al. utilized a hybrid ED/DMRG [249] approach, adopting the parameters $J_1 = 160$ K and $J_2/J_1 = 0.35$ [182]. They showed the scaling of $\delta(T)$ with T and scaling of singlet-triplet gap $\Delta(T)$ with T for $T < T_{SP}$.

4. Normal two leg ladder

A normal ladder consists of two connected chains or legs, where each site of a chain is connected to the nearest neighbor site on the other chain through a rung bond as illustrated in Fig. 9a. The isotropic Heisenberg spin- $\frac{1}{2}$ model on this geometry shows interesting quantum phases which are strikingly different from the ground phases of the MG model of spin- $\frac{1}{2}$ chain. In this system spins are satisfied with their exchange interactions and do not show any sign of frustration if all the exchanges between spins along the legs or rung are of same nature i.e ferromagnetic or antiferromagnetic. Vanadyl pyrophosphate (VO)₂P₂O₇ was considered as first two leg ladder compound as it showed exponentially decaying magnetic susceptibility at low temperature and short range order in neutron scattering indicating finite energy gap in the non-frustrated normal ladder.

The Dagotto et. al. used t-J model on a two leg ladder geometry to understand the superconductive weakly coupled chains, and they notice that in large J limit and at half filling ground state is singlet. The lowest singlet and triplet gap is finite [67, 68] contrary

Table 2. Table for materials related to zigzag ladder with FM nearest neighbor J_1 and AFM next nearest neighbor J_2 .

Materials	References	Model parameters (Kelvin)	Experimental observations
Rb ₂ Mo ₃ Cu ₂ O ₁₂	[191,200,201]	$J_1 = -138, J_2 = 51$ [191,200,201]	Susceptibility, Magnetization and specific heat [191,200]; Nuclear magnetic resonance (NMR) [205,206]
Cs ₂ Mo ₃ Cu ₂ O ₁₂	[200,201]	$J_1 = -93, J_2 = 33$ [200,201]	Susceptibility, Magnetization and specific heat [200,201]; NMR [207]
Li ₂ ZrCuO ₄	[208,209]	$J_1 = -151, J_2 = 35$ [208,209]	Susceptibility, magnetization and specific heat, NMR, INS [208,209]
PbCuSO ₄ (OH) ₂	[197,210–213]	$J_1 = -100, J_2 = 36$ [210]	Susceptibility, magnetization and specific heat, NMR, INS [197,210–213]
LiCuSbO ₄	[189,194,214,215]	$J_1 = -75, J_2 = 34$ [189]	Susceptibility, magnetization and specific heat, NMR, INS [189,194,214,215]
LiCuVO ₄	[216–220]	$J_1 = -18.8, J_2 = 41.3$ [216]; $J_1 = -27.84, J_2 = 39.44$ [217]; $J_1 = -18.45, J_2 = 43.9$ [218]; $J_1 = -76.4, J_2 = 60.32$ [219]	INS [216,217]; Magnetization [218,219]; NMR [220]
NaCuMoO ₄ (OH)	[221,222]	$J_1 = -51, J_2 = 36$ [221]	Susceptibility, magnetization [221]; INS [222]
La ₆ Ca ₈ Cu ₂₄ O ₄₁	[223]	$J_1 = -215, J_2 = 78$ [223]	Susceptibility, magnetization and specific heat, NMR, INS
Ca ₂ Y ₂ Cu ₅ O ₁₀	[223]	$J_1 = -25, J_2 = 55$ [223]	Susceptibility, magnetization and specific heat, NMR, INS
Li ₂ CuO ₂	[223]	$J_1 = -100, J_2 = 62$ [223]	Susceptibility, magnetization and specific heat, NMR, INS
LiCu ₂ O ₂	[224,225]; [226,227]	$J_1 = -138.1, J_2 = 85.9, J_{dc}^b = 37$ [224,225]; $J_1 = -94, J_2 = 167, J_{dc}^b = 4.6$ [226,227]	INS + SWT, Cv(T), $\chi(T)$ [224,225,228]; NMR + LDA [226,227,229]
NaCu ₂ O ₂	[230,231]	$J_1 = -16.4, J_2 = 90$ [230]; $J_1 = -47.5, J_2 = 90.28$ [231]	Susceptibility, magnetization and specific heat, NMR [230,231]
Ba ₃ Cu ₃ In ₄ O ₁₂	[232,233]	$J_1 = -125, J_2 = 0$ [232]; $J_1 = -155, J_2 = 9.3$ [233]	Susceptibility, magnetization and specific heat [232,233]
Ba ₃ Cu ₃ Sc ₄ O ₁₂	[232–234]	$J_1 = -105, J_2 = 0$ [232]; $J_1 = -140, J_2 = 11.2$ [233]	Susceptibility, magnetization and specific heat [232,233], NMR, Muon spin resonance (μ SR) [234]
CuCrO ₄	[235,236]	$J_1 = 54, J_2 = -27$ [235]	Susceptibility, Specific heat [235], Neutron diffraction, μ SR [236]

^b J_{dc} = Interchain interactions.

to the gapless spectrum of an isotropic Heisenberg spin- $\frac{1}{2}$ chain. The study of the two leg ladder systems started with the work of Dagotto, Riera and Scalapino [67], and Dagotto and Moreo [68]. For $t = 0$, the $t - J$ model reduced to an Isotropic Heisenberg model and this model Hamiltonian can be written as

$$\mathcal{H} = \sum_n \sum_{\alpha=1,2} J \vec{S}_{n,\alpha} \cdot \vec{S}_{n+1,\alpha} + \sum_n J_{\perp} \vec{S}_{n,1} \cdot \vec{S}_{n,2}, \quad (10)$$

where first and second term represent the leg and rung terms. α and n correspond to leg number and sites in leg, respectively. The J and $J_{\perp}$ are leg and rung spin exchange strength.

The ground state properties can be tuned by changing the ratio exchange couplings $J/J_{\perp}$ of the two-leg ladder. We will focus our discussion only for antiferromagnetic exchange in the leg and rung exchange can be ferro or anti-ferromagnetic. Three phases can be easily identified as:

Decoupled Chains: When $J_{\perp} = 0$, the ladder reduces to two decoupled spin- $\frac{1}{2}$ chains, resulting in antiferromagnetic ground state with quasi-long range order and spectrum is gapless and highly degenerate. Even a small value of $J_{\perp}$ can open a gap, as shown by Barnes et al. [270].

Haldane phase : For strong ferromagnetic rung exchange $J_{\perp}$, antiferromagnetic leg exchange J the two nearest neighboring spin- $\frac{1}{2}$ from different legs forms a triplet state and the effective spins behave as spin-1 as shown in Fig.9(b). The triplet dimer is three fold degenerate, therefore, each of the triplet dimer can be treated as an effective spin-1. The nearest neighboring effective spin-1 are interacting through antiferromagnetic exchange on a one dimensional chain. The ground state of the system is a valence bond state (VBS) and it shows short range correlation [271]. The spectrum of the system is gaped and singlet-triplet gap is finite. This phase ‘‘Haldane phase’’ is named after the proposer of the phase ‘‘Nobel laureate Prof. Frederick Duncan Michael Haldane’’. [133, 271, 272].

Dimer phase: This phase arises for any finite antiferromagnetic rung exchange coupling $J_{\perp}$, and the ground is a singlet state. Two nearest neighbor spins along the rung form a perfect singlet dimer for dominant antiferromagnetic rung exchange as shown in Fig.9(c). In this limit the wavefunction of i^{th} singlet dimer can be written as $|- \rangle_i = \frac{1}{2}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)_i$ and the ground state of the whole system can be written as product of all singlet dimer wavefunction $|\psi_{gs}\rangle = \prod_i |- \rangle_i$. The lowest spin excitations can be generated by breaking the rung bond which costs energy proportional to $J_{\perp}$ [270]. Schematic representation of the ground state is shown in Fig.9(c). It shows short range magnetic order and correlation

decays exponentially. Therefore, the ground state behaves as spin-liquid state and can also be described by the short-range resonating valence bond (RVB) state [273–277].

Realization of the model: Interestingly, the realization of the two leg ladder started with the early work by Dagotto et al. [67] suggested that $(\text{VO}_2)\text{P}_2\text{O}_7$ was a two-leg ladder compound, distinct from cuprate oxides or halides. Johnston et al. [278] proposed that this material follows a spin- $\frac{1}{2}$ Heisenberg model with a ladder spin configuration and a singlet ground state. Neutron scattering experiments by Eccleston et al. [279] supported this, reporting a energy gap consistent with a two-leg ladder structure. However, later experiments by Garrett et al. [280, 281] revealed that $(\text{VO}_2)\text{P}_2\text{O}_7$ is better described by an alternating Heisenberg antiferromagnetic chain, excluding it from the ladder compound family. Hence, focus shifted to materials containing copper and oxygen.

The first widely accepted material identified as a two-leg ladder is SrCu_2O_3 . This compound contains corner sharing octahedra was proposed by Azuma et al. [282] and also by Hiroi et al. [283] is recognized as an ideal realization of the spin- $\frac{1}{2}$ Heisenberg model on a two-leg ladder as shown in Fig.9. It shows a singlet ground state with a finite energy gap. The structural analysis concluded it as two-leg ladder system. The magnetic properties of the system validates the theoretical predictions for the spin- $\frac{1}{2}$ ladder system, however, hole doping of half-filled Cu systems leads to high temperature superconductor [284, 285]. The strong antiferromagnet Heisenberg coupling among Cu ions is in the ladder and at low temperature weak ferromagnetic inter-ladder exchange sets in. Magnetic susceptibility measurements showed a energy gap of approximately 420K, which roughly agrees with theoretical predictions ($\Delta \approx J/2$) [270], assuming a Heisenberg coupling strength similar to that found in two-dimensional Cu oxides, $J \approx 1300K$.

A large number of materials are synthesized and based on the nature of spin exchange couplings, we can divide materials into at least two categories such as (i) antiferromagnetic leg J and rung $J_{\perp}$, (ii) antiferromagnetic leg J but ferromagnetic rung $J_{\perp}$ etc. For the first case, there are a few materials- SrCu_2O_3 [282, 283, 286–289], $(\text{C}_7\text{H}_{10}\text{N})_2\text{CuBr}_4$ or DIMPY [290–297], $(\text{pipdH})_2\text{CuBr}_4$ or BPCB [298–305], $\text{Cu}_2(\text{C}_5\text{H}_{12}\text{N}_2)_2\text{Cl}_4$ [306–315].

In the SrCu_2O_3 exchange couplings along the legs and rungs of the ladder are antiferromagnetic, and it leads to a large energy gap of approximately 420K, as confirmed by susceptibility data [273, 282]. This was later corroborated by NMR studies, which revealed a slightly larger energy gap $\Delta \approx 680K$ [273, 287, 288], and further theoretical fits by Johnston [286] suggested

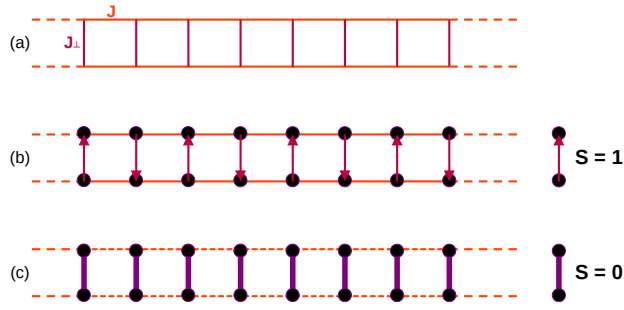


Figure 9. (a) Schematic representation of an isolated normal two leg ladder which has rung exchange, $J_{\perp}$ and leg exchange, J . (b) Schematic of an effective spin-1 chain when rung exchange is ferromagnetic and gives rise to Haldane phase. (c) Schematic representation of the dimer phase, and singlet rung dimers are shown as purple bonds.

a rung coupling is about half of the leg coupling. These findings cemented the connection between theory and experiment in ladder compounds.

In addition to inorganic ladders, metallo-organic compounds can also form spin ladder systems, such as $\text{Cu}_2(\text{C}_5\text{H}_{12}\text{N}_2)_2\text{Cl}_4$. These compounds demonstrate weak exchange interactions due to the long pathway of virtual electron hopping from one Cu atom to another through a large organic ligand. The compound exhibits uneven exchange coupling, with a strong rung coupling ($J_{\perp}/J = 5.5$) [306], leading to a well-defined spin singlet state on the rung bond. Chaboussant et al. (1998b) [316] studied the system in high magnetic field, revealing a rich quantum phase diagram, including quantum critical points and Tomonaga-Luttinger liquid (TLL) behavior of the ground state. These results highlight the theoretical and experimental potential of both organic and inorganic ladder systems, opening new avenues for exploring field-induced quantum transitions and the properties of TLLs in strongly correlated spin systems.

Synthesis of DIMPY ($\text{C}_7\text{H}_{10}\text{N}$) $_2\text{CuBr}_4$ is first compound to demonstrate the strong-leg ladder exchange and it exhibit attractive interactions in the TLL phase [290, 294, 295], as shown through field-induced transitions and low-temperature specific heat measurements. Magnetic field studies on DIMPY revealed that the TLL phase persists between critical fields $H_{c1} \approx 3T$ and $H_{c2} \approx 29T$ [292, 293]. Similarly, for BPCB (pipdH) $_2\text{CuBr}_4$, researchers observed the field-temperature phase diagram, which confirmed the presence of a spin Luttinger liquid phase over a broad temperature range [303]. The extensive study of these ladder systems continues to uncover complex phenomena such as magnetization plateaus [317–319], quantum critical points [299, 320–323], and incommensurate phases [324], making spin ladders a rich area for theoretical and experimental exploration

in quantum magnetism. Table.3 reveals the type of exchange interaction with values of exchanges along the leg and rung of the two leg ladder.

$\text{Ba}_2\text{CuTeO}_6$, a cuprate perovskite, can be described as a tellurium-bridged copper spin- $\frac{1}{2}$ system [336]. The copper spins interact through a Cu-O-Te-O-Cu super-superexchange pathway, forming a novel two-leg spin ladder structure [337]. Inelastic neutron scattering experiments reveal the exchange interactions as $J_{\perp} = J = 109K$ and the interladder coupling as $J' = 49K$ [321].

Ferromagnetic leg and antiferromagnetic rung in a two leg ladder is also popular model to study the exotic quantum phases [338–344]. Weakly anisotropic spin ladders with antiferromagnetically coupled ferromagnetic legs, analyzed within the framework of the non-linear σ approach, exhibit behavior similar to that of an integer-spin antiferromagnetic chain. A phase transition to the antiferromagnetically ordered state occurs only when the anisotropy exceeds a critical threshold [339]. Table 4 provides a list of materials representing such systems.

There are a few studies on ferro- antiferromagnetic legs [345, 346]. The model demonstrates that the partial ferrimagnetic state corresponds to a spontaneously magnetized Tomonaga-Luttinger liquid with incommensurate magnetic correlations, a result validated through DMRG calculations. Using the thermal pure quantum state method, the finite-temperature magnetic susceptibility is computed, indicating a divergence proportional to T^{-2} within the ferrimagnetic phases, similar to the behavior observed in ferromagnetic Heisenberg chains [346].

5. Honeycomb ladder

A honeycomb ladder is structurally similar to a normal ladder, except that it is missing alternate rung bonds. One of the early use of this lattice in condensed matter system was done to understand the effect of electron phonon coupling for a Hubbard model [347]. F. Amiri et. al studied the Heisenberg spin- $\frac{1}{2}$ model on this ladder system [348]. They introduced a parameter α which controls the alternate in rung exchange interaction which can be tuned from ferro to antiferromagnetic as shown in Fig.10. The model Hamiltonian can be written as

$$\mathcal{H} = J \sum_{j=1}^L \sum_{l=1}^2 \vec{S}_{l,j} \cdot \vec{S}_{l,j+1} + J_{\perp} \sum_{j=1}^L [(-1)^j + \alpha] \vec{S}_{1,j} \cdot \vec{S}_{2,j}, \quad (11)$$

where first term indicates the leg spin exchange term and the second term is rung exchange term which alternates. The energy scale is set by $J_{\perp} = 1$ and J can be ferromagnetic or antiferromagnetic.

Table 3. Table for materials related to normal ladder with AFM leg and AFM rung.

Materials	References	Model parameters (Kelvin)	Experimental observations
SrCu ₂ O ₃	[282], [286]	$J \approx 1300$, $J_{\perp} \approx 1300$ [282] (rough agreement with theoretical prediction of energy gap $\Delta \approx J/2$ [270, 325]); $J \approx 2000$, $J_{\perp} \approx 1000$ [286]	Susceptibility + NMR
(C ₇ H ₁₀ N) ₂ CuBr ₄ or DIMPY	[290], [291]	$J = 16.85$, $J_{\perp} = 8.68$ [290]; $J = 19.91$, $J_{\perp} = 9.05$ (from zero field dispersion) [291]; $J = 20.82$, $J_{\perp} = 9.05$ (from energy gap and fermi velocity) [291]	Susceptibility [290] energy gap + fermi velocity + zero field dispersion [291]
(pipdH) ₂ CuBr ₄ or BPCB	[298], [299], [300], [301]	$J_{intra} = 6.64$, $J_{inter} = 0.66$ (considering as double chain of dimers) [298]; $J = 3.8$, $J_{\perp} = 13.3$ [299]; $J = 3.6$, $J_{\perp} = 12.9$ [300, 301]	Susceptibility [298, 299] + perturbation theory [300, 301] + INS [326, 327]
Cu ₂ (C ₅ H ₁₂ N ₂) ₂ Cl ₄	[306], [308], [313]	$J = 2.4$, $J_{\perp} = 13.2$ [306, 313]; $J = 2.79$, $J_{\perp} = 13.96$ [308]	Susceptibility [306] + Series expansion method [308]
Cu ₂ (C ₅ H ₁₂ N ₂) ₂ Br ₄	[328]	$J = 3.5$, $J_{\perp} = 24.8$ [328]	Susceptibility [328]
(Ca,La) ₁₄ Cu ₂₄ O ₄₁	[329]	$J = 1468.8 - 1584$, $J_{\perp} = J$ [329]	Jordan-Wigner fermions [329] + Perturbation theory [329]
BIP-BNO (3,5'-bis;N-tert-butylamino-xyl)-3',5-dibromobiphenyl)	[330], [331]	$J = 25$, $J_{\perp} = 67$ [330]; $J = 14.1$, $J_{\perp} = 65.7$ [331]	Susceptibility [330, 331]
BiCu ₂ PO ₆	[332], [333], [334], [335]	$J = 80$, $J_{\perp} = 72$ [332]	Susceptibility + Specific Heat + First principles method (LDA+DFT) [332] + Raman study [333] + INS [335]

Table 4. Table for materials related to normal ladder with FM Leg and AFM Rung.

Materials	References	Model parameters (Kelvin)	Experimental observations
3-Cl-4-F-V [3-(3-chloro-4-fluorophenyl)-1,5-diphenylverdazyl]	[341]	$J = -13.3$, $J_{\perp} = 7.32$ [341]	Susceptibility [341]
3-Br-4-F-V [3-(3-bromo-4-fluorophenyl)-1,5-diphenylverdazyl]	[343]	$J = -5.1$, $J_{\perp} = 8.4$ [343]	Susceptibility [343]
3-I-4-F-V [3-(3-iodophenyl-4-fluorophenyl)-1,5-diphenylverdazyl]	[343]	$J = -9.4$, $J_{\perp} = 7.8$ [343]	Susceptibility [343]

F. Amiri et. al noted that ferro or ferrimagnetic state for ferromagnetic legs and negative α , whereas for positive α ground state is a half ferro state [348]. For the antiferromagnetic leg couplings, various type of short-range ground states including Haldane phase and Haldane dimer phase along the rung as shown in Fig.10(b), are suggested for negative α [348]. These ground states may emerge by tuning the rung exchange interaction. The next-nearest-neighbor diagonal dimer (NNND-Dimer) phase appears in the ground state for antiferromagnetic J and positive α , as shown in Fig.11. For $J_{\perp}(\alpha-1) = 0$ and $J_{\perp}(\alpha+1) = J'$ in Hamiltonian 11, the model transforms to the original structure of honeycomb ladder.

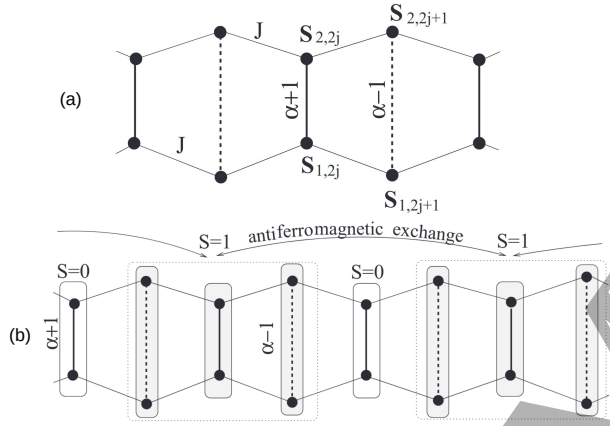


Figure 10. (a) Schematic of the two leg honeycomb-ladder model. (b) Illustration of a possible ground-state configuration in the Haldane-dimer phase, achievable for $\alpha \approx -1$ and $0 < J \ll 1$. Spins within open rectangles represent approximate rung-singlet states, while those within shaded rectangles correspond to rung-triplet states. Six spins enclosed by the dotted rectangle form effective $S = 1$ spins, which are linked via intermediate singlets to create an effective Haldane chain. In the mixed-diamond chain, the singlets depicted here are exact eigenstates and do not facilitate any exchange interactions between the effective $S = 1$ spins. Figure adapted from Ref. [348].

Realization of the model: Y. Y. Zhu et al first synthesized a two leg honeycomb ladder material [349]. In the four azide copper coordination compound $[\text{Cu}_2\text{L}_1(\text{N}_3)_4]_n$ [349], copper ions with spin- $\frac{1}{2}$ are coupled via N_3 bridges, forming a two-leg honeycomb spin ladder. The spin exchange interactions were calculated by Y. Zhao et al. [350], revealing uniform interactions with values $J = 0.17K$ (along the leg) and $J' = -2.39K$ (along the rung) [350].

6. Weakly coupled dimer spins on a ladder geometry

Two antiferromagnetically coupled spin- $\frac{1}{2}$ form a singlet dimers and shows large singlet-triplet gap and the excitation is called triplon. The singlet

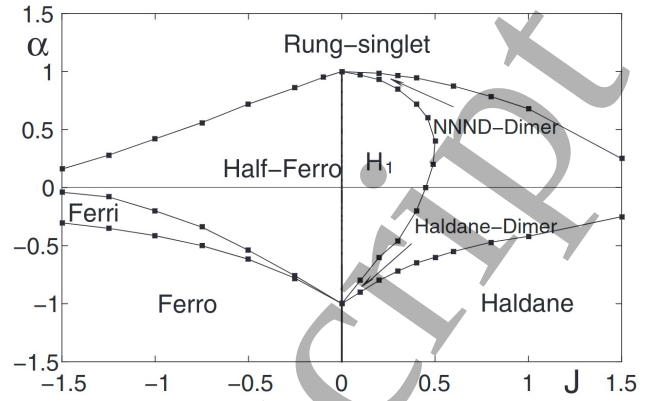


Figure 11. Numerical ground-state phase diagram of the honeycomb-ladder model in the parameter plane (J, α) . NNND-dimer refers to the next-nearest-neighbor diagonal dimer phase, where dimers are formed along the diagonals connecting next-nearest-neighbor spins on odd rungs. Close to $\alpha = -1$ and for $0 < J \ll 1$, additional phases may appear before the transition from the Haldane-dimer phase to the Haldane phase. In dimer phases, the ground states exhibit double degeneracy in the thermodynamic limit. Phase diagram adapted from Ref. [348].

dimer ground state can be realized in various spin- $\frac{1}{2}$ systems like spin-peierls or dimerized spin chain [250, 351–354], finite leg ladders [348, 355, 356], frustrated one dimensional spin- $\frac{1}{2}$ [118, 357–359] systems etc. The field-induced phases are also observed in several high spin compounds but the quantum effects are more predominant in systems with spin- $\frac{1}{2}$ [360, 361]. In many systems spin of the spin dimers can have finite exchange coupling with neighboring spin dimers as in Sashtry-Sutherland (SS) model in two dimension [362, 363] or ladder like structure [364]. SS model in two dimension have orthogonal dimers and these are coupled with each other with antiferromagnetic [362, 365–367] or ferromagnetic [364] exchange interaction. The coupled dimers shows many interesting properties and ground state properties depends on the exchange coupling between the dimers. Many of these system shows interesting properties of Bose-Einstein condensation at finite axial magnetic field [368–370]. The presence of interdimer couplings results in a reduction of the gap value, rendering these compounds well-suited for high-field studies [371, 372].

A large number of coupled dimer systems have been synthesized, such as those reported in [373–375]. P. Biswal et al. [376] synthesized the two-leg ladder compounds $(\text{C}_4\text{H}_{14}\text{N}_2)\text{Cu}_2\text{X}_6$ ($X = \text{Cl}, \text{Br}$), which crystallize in an orthorhombic structure with the space group $Pnma$. The crystal structure consists of Cu^{2+} dimers arranged parallel to each other, forming a zigzag two-leg ladder-like arrangement, as depicted in Fig. 12. The effective model Hamiltonian for the ladder, with weak interladder coupling as illustrated in Fig. 12, can

be expressed as:

$$\begin{aligned} \mathcal{H} = & \sum_{i=1}^{N/2} J_0 \vec{S}_{i,A} \cdot \vec{S}_{i,B} + \sum_{i=1}^{N/2} J' \vec{S}_{i,B} \cdot \vec{S}_{i+1,B} \\ & + \sum_{i=1}^{N/2} J'' (\vec{S}_{i,B} \cdot \vec{S}_{i+1,A} + \vec{S}_{i,A} \cdot \vec{S}_{i+1,B}) \\ & - H \sum_{i=1}^{N/2} (S_{i,A}^z + S_{i,B}^z), \end{aligned} \quad (12)$$

where $\vec{S}_{i,A}$ and $\vec{S}_{i,B}$ indicate the spin vectors for the sub-lattices A and B, respectively of the i^{th} unit cell. In Eq.(12), the first term represents the intradimer coupling J_0 while the second and third terms represent the inter-dimer coupling J' and J'' between the neighboring dimers along the diagonal and the leg, respectively.

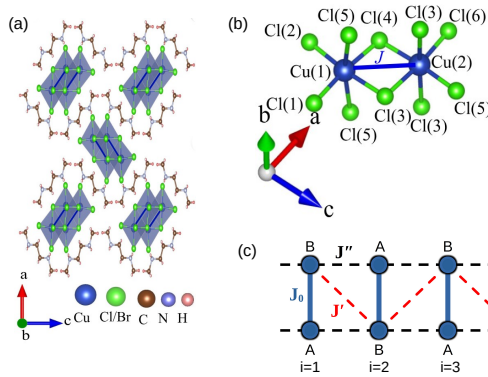


Figure 12. (a) Three-dimensional view of the crystal structure of $(C_4H_{14}N_2)Cu_2X_6$ ($X = Cl, Br$) projected onto the ac -plane. (b) The Cu_2X_{10} ($X = Cl, Br$) dimer unit showing the intradimer exchange coupling J_0 (Figs. (a) and (b) are adapted from P. Biswal et al. [376]). (c) Lattice structure: A and B represent two sublattices of the ladder structure. J_0 and J'' correspond to the rung and leg couplings, respectively, while J' denotes the diagonal coupling.

Experimental Realization : P. Biswal et al. synthesized two copper halides [376], $(C_4H_{14}N_2)Cu_2X_6$ ($X = Cl$ and Br), which were theoretically modeled using the Hamiltonian in Eq. (12) for the two sublattices, A and B, corresponding to two types of copper atoms, as shown in Fig. 12b. They determined the intradimer exchange parameter values as $J_0 = 116$ K, and the two interdimer exchange parameters as $J' = 23.2$ K and $J'' = 18.6$ K for $(C_4H_{14}N_2)Cu_2Cl_6$, and $J_0 = 300$ K, $J' = 90$ K, and $J'' = 105$ K for $(C_4H_{14}N_2)Cu_2Br_6$, with $g = 2.06$ in both cases. Although both compounds share the same crystal symmetry, the relatively stronger exchange couplings in the Br compound compared to the Cl compound are attributed to the larger ionic size of Br^- and its more diffuse p -orbitals, which enhance the coupling between Cu^{2+} ions. Additionally, the study identified

large energy gaps of 78.9 K and 181.2 K for the Cl and Br compounds, respectively, indicating that the robust singlet ground state remains unaffected by experimentally accessible magnetic fields.

Metal-organic compounds are particularly well-suited for magnetic field-dependent studies due to their tunable interdimer and intradimer exchange couplings. This tunability allows the energy gap and ground state properties to be adjusted by modifying synthesis conditions and altering the ligand [377]. For example, the isolated spin- $\frac{1}{2}$ dimers in the metal-organic compound $Cu_2(IPA)_2(DMF)(H_2O)$ [374] exhibit a large energy gap of approximately 414 K, whereas isolated dimers in $Cu_2Mg_2(CO_3)(OH)_6 \cdot 2H_2O$ [375] have a significantly smaller energy gap of around 7 K. Recently, numerous new magnetic materials consisting of antiferromagnetic spin dimers have been identified, with a variety of magnetic excitations observed under external magnetic fields for different interdimer exchange networks [378–383]. However, as these compounds do not form two-leg ladders, they fall outside the scope of this article.

7. Diamond ladder

The Heisenberg diamond ladder, characterized by diamond-shaped unit cells arranged along the chain, has attracted significant attention in both experimental and theoretical studies [384–398].

Ishii et al. [387] reported experimental findings for a trimerized spin- $\frac{1}{2}$ quantum spin chain, $Cu_3Cl_6(H_2O)_2 \cdot 2H_8C_4SO_2$. They also proposed a model Hamiltonian based on structural analysis, as shown in Fig. 13.

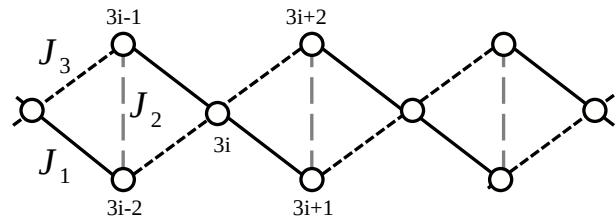


Figure 13. Schematic representation of the diamond chain. The exchange interactions, J_1 , J_2 , and J_3 , are indicated, all of which are antiferromagnetic in nature. J_1 set the energy scale in the diamond lattice model and set $\tilde{J}_2 \equiv \frac{J_2}{J_1}$ and $\tilde{J}_3 \equiv \frac{J_3}{J_1}$.

The model Hamiltonian of the system is expressed

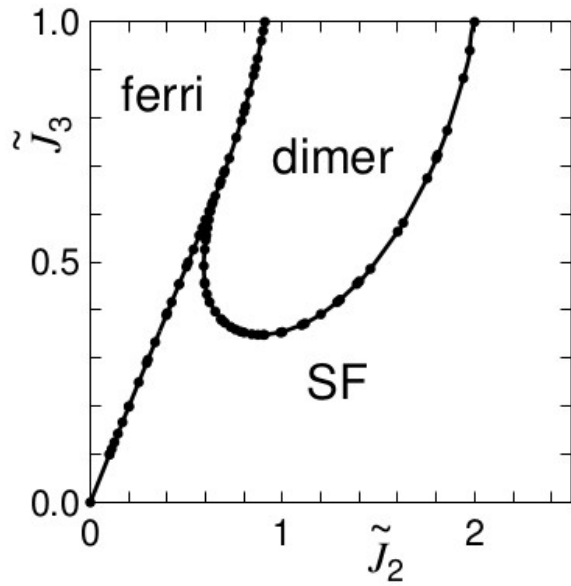


Figure 14. Ground-state phase diagram of the $S = \frac{1}{2}$ distorted diamond chain model in $\tilde{J}_2 - \tilde{J}_3$ parameter space, highlighting three distinct phases: the ferrimagnetic (ferri) phase, the dimer phase, and the spin fluid (SF) phase. The Fig. is reproduced from Ref. [401].

as:

$$\begin{aligned} \mathcal{H} = & J_1 \sum_j \left(\vec{S}_{3i-1} \cdot \vec{S}_{3i} + \vec{S}_{3i} \cdot \vec{S}_{3i+1} \right) \\ & + J_2 \sum_j \vec{S}_{3i+1} \cdot \vec{S}_{3i+2} \\ & + J_3 \sum_j \left(\vec{S}_{3i-2} \cdot \vec{S}_{3i} + \vec{S}_{3i} \cdot \vec{S}_{3i+2} \right), \end{aligned} \quad (13)$$

where the three spins $\vec{S}_{3i-1}$, $\vec{S}_{3i}$, and $\vec{S}_{3i+1}$ form a trimer. All coupling constants are antiferromagnetic. For convenience, we set J_1 to define the energy scale in the diamond lattice model, with $\tilde{J}_2 \equiv \frac{J_2}{J_1}$ and $\tilde{J}_3 \equiv \frac{J_3}{J_1}$.

This model has been referred to as the “distorted diamond (DD) chain model” [392]. Okamoto et al. [399] investigated the ground state properties of the spin- $\frac{1}{2}$ DD chain model at zero field. Subsequent theoretical studies on the spin- $\frac{1}{2}$ DD chain were conducted by Okamoto et al. [395, 400], Sano and Takano [394], and Honecker and Lauchli [397].

The spin- $\frac{1}{2}$ double-chain (DD) model has been realized in various materials, such as $A_3\text{Cu}_3(\text{PO}_4)_4$ ($A = \text{Ca}, \text{Sr}$) [384, 385], $\text{Bi}_4\text{Cu}_3\text{V}_2\text{O}_{14}$ [386], and azurite $\text{Cu}_3(\text{CO}_3)_2(\text{OH})_2$ [389]. These compounds exhibit a $\frac{1}{3}$ -magnetization plateau [389, 390].

Quantum phases: The quantum phase diagram of the model in the $\tilde{J}_2 - \tilde{J}_3$ parameter space is shown in Fig. 14. This phase diagram features three distinct quantum phases: the ferrimagnetic (ferri), dimer, and spin fluid (SF) phases. The total spin magnitude, S_{tot} ,

is defined as $S_{\text{tot}} = \sum_{j=1}^N S_j$, with $S_{\text{tot}}^2 = S_{\text{tot}}(S_{\text{tot}} + 1)$. In the ferrimagnetic phase, the total spin is $S_{\text{tot}} = \frac{N}{6}$, while in both the dimer and SF phases, the ground state spin is $S_{\text{tot}} = 0$.

In the dimer phase, a finite energy gap separates the doubly degenerate ground state from the lowest triplet state. In contrast, the spin-fluid phase is characterized by the absence of such an energy gap. For $J_2 = 0$, the ground state is a ferrimagnetic state with total spin $S_{\text{tot}} = N/6$, as predicted by the Lieb–Mattis theorem [402]. Pictorial representations of the ferrimagnetic ground state are shown in Fig. 15, corresponding to the cases $J_1 \sim J_3$ and $J_1 \gg J_2, J_3$, respectively. The ellipses in Fig. 15 highlight the effective three-spin clusters that arise when the interaction J_1 significantly exceeds the strengths of J_2 and J_3 .

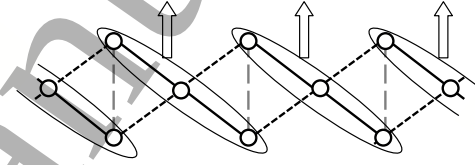


Figure 15. Physical representation of the ferrimagnetic ground state in the $S = \frac{1}{2}$ DD chain model. The ellipses with spins illustrate the three-spin clusters in the ϕ_1 state.

The $S_i^z = 1/2$ and $-1/2$ states of the i th spin are denoted as $\uparrow_i$ and $\downarrow_i$, respectively. When $J_2 = J_3 = 0$, the two degenerate ground-state wave functions of a cluster of three spins are represented as follows:

$$\begin{aligned} |\phi_1\rangle = & \frac{1}{\sqrt{6}} (|\uparrow_{3i-1}\uparrow_{3i}\downarrow_{3i+1}\rangle - 2|\uparrow_{3i-1}\downarrow_{3i}\uparrow_{3i+1}\rangle \\ & + |\downarrow_{3i-1}\uparrow_{3i}\uparrow_{3i+1}\rangle) \end{aligned} \quad (14)$$

and

$$\begin{aligned} |\phi_2\rangle = & \frac{1}{\sqrt{6}} (|\downarrow_{3i-1}\downarrow_{3i}\uparrow_{3i+1}\rangle - 2|\downarrow_{3i-1}\uparrow_{3i}\downarrow_{3i+1}\rangle \\ & + |\uparrow_{3i-1}\downarrow_{3i}\downarrow_{3i+1}\rangle) \end{aligned} \quad (15)$$

A quantum phase transition between the ferrimagnetic and spin-fluid phases occurs at $\tilde{J}_3 = \tilde{J}_2$, in proximity to the point $(\tilde{J}_3, \tilde{J}_2) = (0, 0)$. This transition is characterized as being of the first order.

The quantum phase transition between the ferrimagnetic and dimer phases is also of first order. When $J_3 = 0$, the DD chain reduces to a J_1 - J_1 - J_2 trimerized antiferromagnetic chain, whose ground state is a spin fluid state. Specifically, for $J_1 = J_2$, the system further simplifies to a uniform antiferromagnetic chain.

The dimer phase emerges due to frustration, leading to doubly degenerate due to the spontaneous

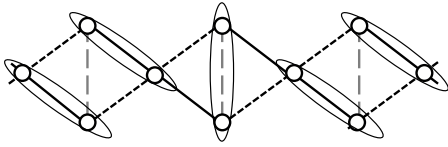


Figure 16. Physical representation of the dimer ground state for the $S = \frac{1}{2}$ DD chain model. Two spins in an ellipse effectively form a singlet dimer pair.

breaking of the translational symmetry. The doubly degenerate ground states are pictorially presented in Figs. 16 (a) and (b). The quantum phase transition between spin fluid to dimer is essentially same as that in the spin- $\frac{1}{2}$ $J_1 - J_2$ model on a chain.

Experimental Realization: Kikuchi et al. investigated the spin- $\frac{1}{2}$ compound $\text{Cu}_3(\text{CO}_3)_2(\text{OH})_2$, known as azurite, where copper atoms are arranged in quasi-one-dimensional diamond chains composed of corner-sharing rhombic plaquettes [390]. The chain exhibits dimerization along the rungs, driven by the dominant antiferromagnetic (AFM) exchange interaction J_2 between nearest-neighbor Cu(2) sites, with J_2 values reported to range from 2.1 meV to 3.8 meV, depending on the model used [403, 404]. In addition to J_2 , the system includes asymmetric Cu(1)-Cu(2) couplings, where $J_1 \approx 0.47J_2$ and $J_3 \approx 0.21J_2$. A direct exchange interaction, $J_4 \approx 0.14J_2$, is also considered in some studies [403, 405]. A notable characteristic of this system is the appearance of a $1/3$ -magnetization plateau phase [390], associated with the polarization of monomers under an external magnetic field, occurring between the critical fields $B_{c1} = 11$ T and $B_{c2} = 30$ T.

The experimental observations at high magnetic fields are accurately captured by results from exact diagonalization and Dynamical-DMRG (DDMRG) calculations [403, 404]. Honecker et al. calculated the dynamic structure factor at zero magnetic field using DDMRG on a chain [404]. The spectrum reveals a gapless two-spinon continuum at low energies ($E \leq 2$ meV), distinctly separated from a broad band of higher-energy dimer excitations. These results qualitatively align with earlier neutron spectroscopy measurements [405].

Another material of interest is $\text{Cu}_3\text{Cl}_6(\text{H}_2\text{O})_2 \cdot 2\text{H}_8\text{C}_4\text{SO}_2$. Magnetization and susceptibility measurements conducted by Ishii et al. [387] revealed the existence of a energy gap of approximately 3.9 T (roughly equivalent to 5.5 K) in this system. In addition to the energy gap, the system likely exhibits a plateau at one-third of the saturation magnetization. Fits of the high-temperature tail of the susceptibility, calculated from the model, to the experimentally measured susceptibility on $\text{Cu}_3\text{Cl}_6(\text{H}_2\text{O})_2 \cdot 2\text{H}_8\text{C}_4\text{SO}_2$ [387] yield

the following parameter estimates: $J_2 = 250 \text{ K} \pm 40 \text{ K}$, $J_1 = -260 \text{ K} \pm 50 \text{ K}$, and $J_3 = -40 \text{ K} \pm 30 \text{ K}$.

8. Skewed ladder

In the first section, we explained how a spin model with nearest and next-nearest neighbor interactions on a one-dimensional chain can be mapped to interacting spins on zigzag ladders, as illustrated in Fig.17. Notably, a wide variety of ladder geometries, known as n/m skewed ladders, can be systematically constructed by selectively removing specific bonds from a standard zigzag spin ladder. In this context, the integers n and m denote the number of sites forming two alternating ring types along the ladder structure. The term “skewed ladder” refers to the tilted orientation of the rung bonds, which are no longer perpendicular but instead slanted due to this bond deletion pattern [406, 407]. This procedure results in a periodic arrangement of fused polygonal plaquettes, typically n - and m -membered rings, creating a rich class of geometrically frustrated structures. Examples of such configurations include $5/7$, $3/4$, $3/5$, $5/5$, and $3/3$ skewed ladders, each with distinct symmetry and connectivity features [406].

5/7-Skewed ladder: This structure is inspired by fused azulene, which consists of alternating 5- and 7-membered carbon rings along a chain. Thomas et al. first studied this structure, demonstrating that the ground state is ferrimagnetic [407]. Subsequently, the isotropic antiferromagnetic Heisenberg model Hamiltonian for this ladder has been extensively studied, and can be expressed for an unit cell (shown in Fig.17(d)) as:

$$\mathcal{H} = J_1 (\vec{S}_1 \cdot \vec{S}_2 + \vec{S}_4 \cdot \vec{S}_5) + J_2 (\vec{S}_1 \cdot \vec{S}_3 + \vec{S}_2 \cdot \vec{S}_4 + \vec{S}_3 \cdot \vec{S}_5 + \vec{S}_4 \cdot \vec{S}_6 + \vec{S}_6 \cdot \vec{S}_8 + \vec{S}_5 \cdot \vec{S}_7) \quad (16)$$

where the first term represents the rung exchange and the second term represents the exchange along the legs. Here, J_2 sets the energy scale and is equal to one.

The magnetic properties of the ground state are influenced by variations in J_1 . For small values of J_1 , the ground state is a singlet phase, with spins on each leg aligned antiferromagnetically. The rung exchange J_1 induces frustration within the system and opens a energy gap. Fig.18 shows the excitation gap as a function of J_1 for a system size of $N = 24$ sites with periodic boundary conditions (PBC) [406]. As we discussed earlier in Section 3, according to the LSM theorem [121, 122, 408], a spin system with translational and spin rotation symmetry and half-integer spin per unit cell cannot possess a gapped symmetric ground state devoid of fractionalized excitations [121, 122]. In a finite system, these gaps remain finite, and the lowest excitation can be either triplet or singlet which

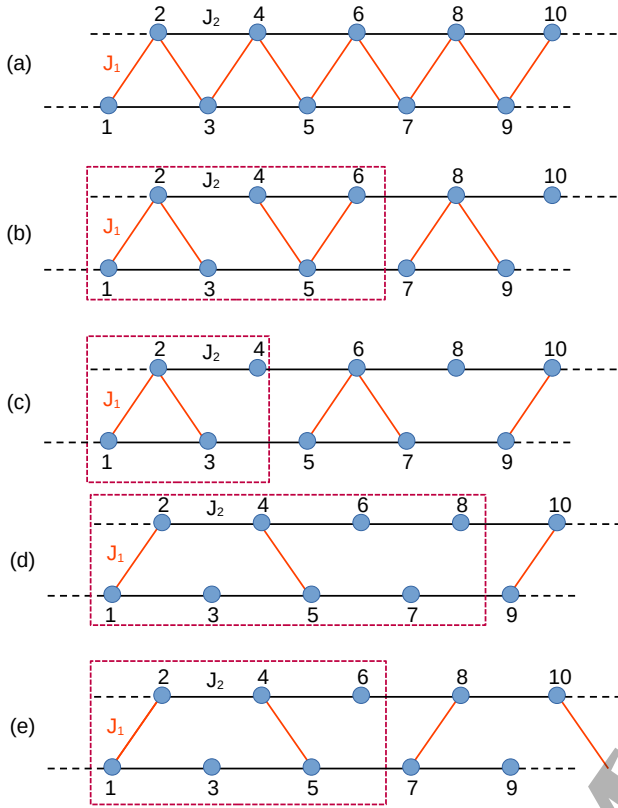


Figure 17. Structure of various types of skewed ladders derived from a zigzag ladder. (a) Zigzag ladder with nearest-neighbor coupling J_1 and next-nearest-neighbor coupling J_2 . (b) 3/4 skewed ladder. (c) 3/5 skewed ladder. (d) 5/7 skewed ladder. (e) 5/5 skewed ladder. Dashed rectangle box represents the unit cell of each ladder structure.

may gapless or doubly degenerate respectively in the thermodynamic limit.

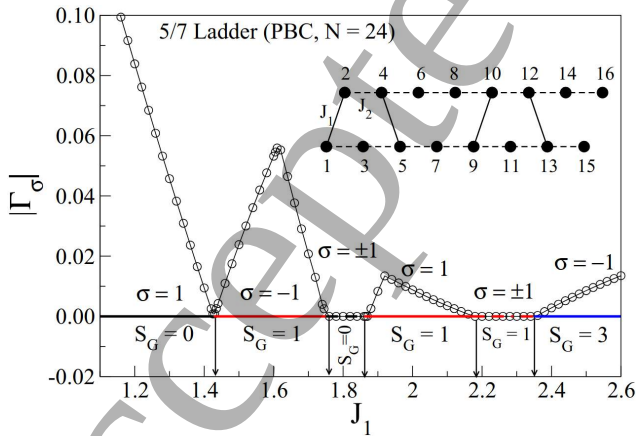


Figure 18. Excitation $\Gamma_\sigma = E_0(S_G, \sigma = -1, N) - E_0(S_G, \sigma = 1, N)$ for a 5/7 ladder with periodic boundary conditions (PBC) and 24 spins; inset shows two unit cells. The quantum phases are characterized by spin S_G and inversion symmetry σ . Figure is adapted from [406].

In this system for size $N = 24$ the singlet ground state extends up to $J_1 = 1.43$ and ground state goes to a triplet state thereafter. The transition from singlet to finite total spin S_G occurs around $J_1 \approx 0.85$ in larger systems, as illustrated in Fig. 18 [406]. A narrow window exists where the ground state remains in the singlet phase, specifically $1.75 < J_1 < 2.18$, as shown in Fig.10 of ref. [406]. For $J_1 > 2.18$, the ground state transitions to a high spin state ($S_G > 0$), and for $J_1 > 2.35$, it enters a magnetic state where $S_G = N/8 = n$, with N being the number of sites and n the number of unit cells in the system. This system also exhibits an intriguing vector chiral phase, as shown in Fig.10 of ref. [406]. For $2.18 < J < 2.35$, the ground state is double degenerate, with these two states related by inversion symmetry and a total spin $S_G = 3$. The broken inversion and spin parity symmetry of the ground state lead to a finite spin current in the system [406]. At finite magnetic fields, the ground state reveals several interesting plateau phases [409].

3/4-Skewed ladder: This is one of the simplest ladder structures, formed by alternately fusing 3- and 4-member rings, as shown in Fig. 17(b). This system is topologically similar to a diamond lattice, featuring additional next-nearest neighbor interactions [406]. It has six spins per unit cell, and the model Hamiltonian for the unit cell (shown in Fig.17(b)) can be expressed as:

$$\mathcal{H} = J_1(\vec{S}_1 \cdot \vec{S}_2 + \vec{S}_2 \cdot \vec{S}_3 + \vec{S}_4 \cdot \vec{S}_5 + \vec{S}_5 \cdot \vec{S}_6) + J_2(\vec{S}_2 \cdot \vec{S}_4 + \vec{S}_3 \cdot \vec{S}_5) \quad (17)$$

where J_1 and J_2 represent nearest neighbor spin exchanges along the rungs and legs, respectively. The leg exchange J_2 is set to one while J_1 is varied. Both exchanges are isotropic and antiferromagnetic.

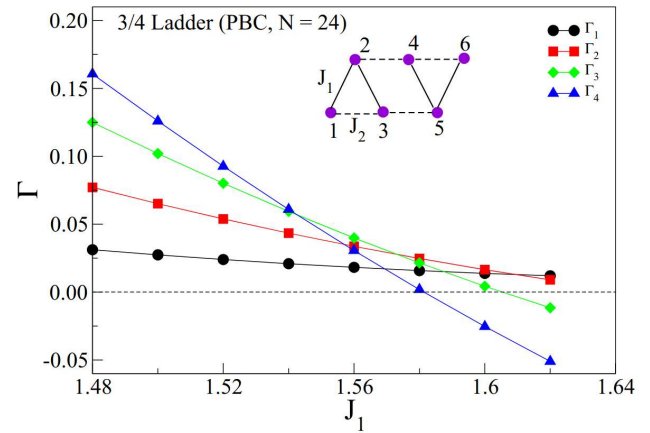


Figure 19. Energy differences Γ_n in Eq.(18) for sectors with total spin S as a function of J_1 in a 3/4 ladder with periodic boundary conditions (PBC) and 24 spins; inset shows two unit cells of the 3/4 ladder under PBC. Figure is adapted from [406].

To determine the ground state spin configuration, we calculate the energy gap between the singlet state and the lowest state in total spin S , denoted as Γ_n , defined as:

$$\Gamma_n = E_0(S, N) - E_0(0, N), \quad (18)$$

where $E_0(S, N)$ and $E_0(0, N)$ are the lowest state energies in the total spin S and singlet sectors, respectively. In Fig. 19, we plot Γ_n for a system size of $N = 24$ sites or four unit cells. We observe that Γ_4 becomes negative for $J_1 > 1.56$, indicating a transition from the singlet to a high spin state. For larger $J_1 > 1.57$, the ground state spin corresponds to the number of unit cells. The high spin ground state can be understood as arising from the spin- $\frac{1}{2}$ on each triangle coupling ferromagnetically with their nearest neighboring spins. The quantum phase of spin-1 on a 3/4 ladder exhibits numerous interesting plateau phases [410]. Investigations into the effects of Coulomb interactions within the Hubbard model on the 3/4 ladder geometry reveal many intriguing features, including high spin insulating states, Tomonaga-Luttinger liquid phases, and other phenomena [411].

5/5 Skewed ladder: This ladder consists of two inverted five-membered rings, as shown in Fig.17(e). There are two J_1 exchanges for every six J_2 exchanges, and the model Hamiltonian for the unit cell (shown in Fig.17(e)) can be expressed as [406]:

$$\mathcal{H} = J_1(\vec{S}_1 \cdot \vec{S}_2 + \vec{S}_4 \cdot \vec{S}_5) + J_2(\vec{S}_1 \cdot \vec{S}_3 + \vec{S}_2 \cdot \vec{S}_4 + \vec{S}_3 \cdot \vec{S}_5 + \vec{S}_4 \cdot \vec{S}_6) \quad (19)$$

where the first term represents rung exchanges, and the second term accounts for leg exchanges. J_2 is set to one.

The ground state of this system remains in the singlet state across the entire parameter regime of J_1 and J_2 . For large J_1 , the spin at the third site becomes localized, as it is the only site not connected by J_1 exchange. In this limit, all other sites form strong singlet dimers through rung bonds, while the spin at the third site behaves as an isolated effective spin- $\frac{1}{2}$. These isolated spins form effective antiferromagnetic exchanges with their corresponding isolated nearest neighbor sites, with the effective exchange proportional to J_2^2/J_1 .

The ground state of the system goes to a paramagnetic phase in the limit of infinite J_1 , due to the formation of perfect dimers, resulting in the third spin acting as an isolated spin. In the thermodynamic limit at large J_1 , it behaves as HAF for the localized spins at the third site; the ground state is a singlet and the spin correlation shows quasi-long-range order.

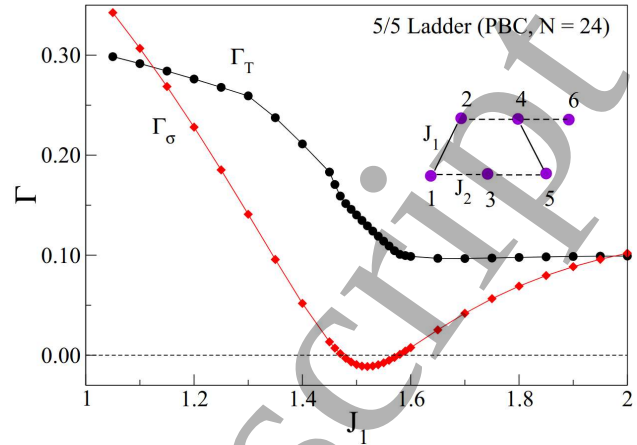


Figure 20. Excitation energies for a 5/5 ladder with periodic boundary conditions (PBC) and 24 spins: Γ_T to the ground-state triplet and Γ_σ to the lowest singlet with inversion symmetry opposite to that of the singlet ground state; inset illustrates two unit cells. Figure is adapted from [406].

The lowest excitation is a gapless triplet state. To characterize the ground state phases, we calculate the two lowest excitations: the single-triplet gap E_m and the lowest singlet excitation E_σ as functions of J_1 , as depicted in Fig. 20 for $N = 24$ sites. For small J_1 , the lowest excitation is E_m ; however, there is a crossover from E_m to E_σ at $J_1 = 1.13$, with another crossover between E_σ and E_m occurring at $J_1 = 2$ for the finite system. For $1.13 < J_1 < 2$, the lowest excitation is a singlet, and the relationship between the lowest singlet excitation and the ground state energy is governed by inversion symmetry.

Now, the question arises: how can these skewed ladders be realized? Recently, various theoretical efforts have been made to address this. Short oligomers of fused azulene (up to six azulene units) have been studied using unrestricted density functional theory and spin models, which indicated a triplet ground state for longer oligomers [412]. Additionally, Rano et al. have investigated a related system, namely fused acene-azulene structures, using ab initio methods in their pursuit of triplet ground states [413]. There have been numerous theoretical attempts to design magnetic ground states for hydrocarbon-based systems analogous to skewed ladders [413–416].

Moreover, it is possible to fabricate such ladders as defects in graphene layers or at the grain boundaries in graphene [417–419]. In these structures, the electrons in the $2p_z$ orbital of the carbon atoms form a strongly correlated band that can be modeled using a long-range interacting framework such as the Pariser-Parr-Pople (PPP) model [420, 421]. This PPP model can be transformed into an isotropic spin- $\frac{1}{2}$ Heisenberg model with antiferromagnetic exchange interactions.

Some of the magnetic properties of the ground

state have been observed in 3/4 skewed distorted azurite materials, such as $\text{Cu}_3(\text{CO}_3)_2(\text{OH})_2$ [389, 422], as well as in the weakly interacting trimer $\text{Na}_2\text{Cu}_3\text{Ge}_4\text{O}_{12}$ [66, 423] at finite magnetic field. The organic compound 2-[3',5'-bis(N-tert-butylaminoxyl)phenyl]-4,4,5,5-tetramethyl-4,5-dihydro-1H-imidazol-1-oxyl 3-oxide, or PNNBNO, which forms an AB2 structure, also exhibits ferrimagnetism at low temperatures [424–426] which is similar to that of 3/4 ladder.

9. Kitaev-Heisenberg model on spin- $\frac{1}{2}$ ladder

The Kitaev model, introduced by A. Kitaev in 2006, is a prominent model that describes a highly frustrated spin system exhibiting exotic phases of matter, particularly in the context of quantum spin liquids (QSLs). The model is defined on a two-dimensional honeycomb lattice and features anisotropic spin exchange interactions, where the interactions between nearest-neighbor sites take three distinct forms: $S^x S^x$, $S^y S^y$, or $S^z S^z$, depending on the bond direction within the honeycomb lattice., as shown in Fig. 21.

The Kitaev model Hamiltonian on the honeycomb lattice is given by [427]:

$$\mathcal{H}_{\text{Kitaev}} = -K_x \sum_{\text{x-bonds}} S_i^x S_j^x - K_y \sum_{\text{y-bonds}} S_i^y S_j^y - K_z \sum_{\text{z-bonds}} S_i^z S_j^z. \quad (20)$$

This model is exactly solvable by expressing spin operators in terms of Majorana fermions, which allows for the exact determination of the ground-state properties. For the isotropic Kitaev model, the ground state is a gapless Kitaev spin-liquid state, characterized by gapless Majorana fermion excitations, which manifest as two Dirac cones [428]. The spin-spin correlations in this gapless Kitaev spin-liquid are short-range, with non-zero values appearing only for nearest-neighbor pairs [429].

Kitaev interaction in real materials: The experimental realization of the Kitaev model and the spin liquid phase remains an open question in condensed matter physics. The concept of realizing the Kitaev model in real materials was proposed by Jackeli and Khaliullin in 2009 [430]. These types of magnetic systems are prominently found in iridium-based materials, such as A_2IrO_3 ($\text{A} = \text{Na}, \text{Li}$), in which iridium atoms form a honeycomb lattice structure [431–433]. In these materials, the octahedral coordination of ligands induces a crystal field that splits the Ir d -orbitals into three t_{2g} and two e_g levels. five d electrons occupy the t_{2g} orbitals with an effective orbital moment $l = 1$. Strong spin-orbit coupling further splits the t_{2g} orbitals into fully

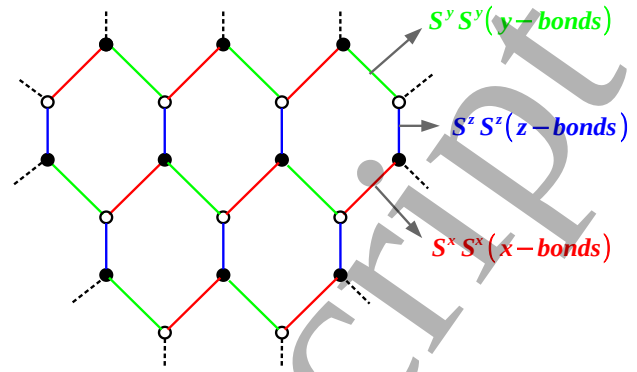


Figure 21. Schematic representation of the Kitaev model on a honeycomb lattice. The indices x, y, z correspond to the three different types of bonds: x-bond, y-bond, and z-bond, respectively.

$j = 3/2$ and half filled $j = 1/2$ band. Strong electronic correlations drive the system into a $j = 1/2$ Mott insulating state. This gives rise to orbital-dependent, directionally anisotropic spin exchanges, known as Kitaev interactions [434]. However, the Kitaev-Heisenberg (KH) model, which includes residual nearest-neighbor Heisenberg interactions, provides a more accurate description of the magnetic properties of these materials [431]. Similar Hamiltonians have also been proposed for other materials, including α - RuCl_3 [434–439], β - Li_2IrO_3 [434, 438–441], and γ - Li_2IrO_3 [434, 442].

Kitaev-Heisenberg ladder: We now examine the Kitaev-Heisenberg (KH) model on a two-leg ladder, as shown in Fig. 22. While the original Kitaev spin liquid (KSL) was defined on the honeycomb lattice, it is well-known that the Kitaev interaction can exhibit interesting properties on any three-coordinated lattice. In this context, the KH model on the ladder, shown in Fig. 22(a) and referred to as the "KH ladder," naturally satisfies the necessary geometric conditions due to the presence of the z bonds.

The KH ladder can also be derived from a brick-wall lattice, as shown in Fig. 22(b), which is geometrically equivalent to the honeycomb lattice. By cutting along the grey line and folding the z bonds towards the center, the KH ladder configuration in Fig. 22(a) is recovered. This construction offers a way to better understand the fundamental properties of the honeycomb-lattice KH model by studying the KH ladder.

The Kitaev-Heisenberg (KH) Hamiltonian defined on a two-leg ladder is given by:

$$\mathcal{H} = K \sum_{\langle i,j \rangle, \gamma=x,y,z} S_i^\gamma S_j^\gamma + J \sum_{\langle i,j \rangle} \vec{S}_i \cdot \vec{S}_j \quad (21)$$

where, $S = \frac{1}{2}$, $\langle i,j \rangle$ represents site indices defined on nearest neighbor (NN) bonds, and $\gamma = x, y$, or z

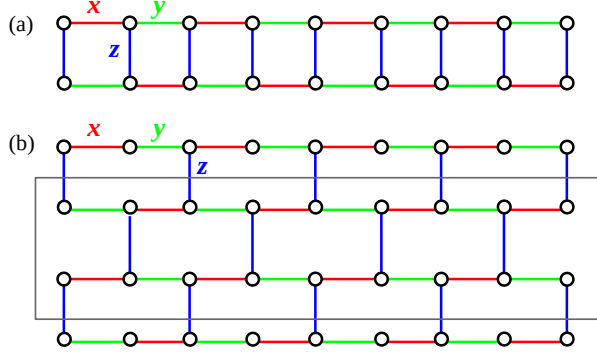


Figure 22. (a) Schematic lattice structure of the KH model on a two-leg ladder. (b) Schematic structure of the KH model on a brickwall lattice, which is geometrically equivalent to the honeycomb lattice. The grey rectangle highlights a cutout that enables the mapping to the ladder in (a). The indices x , y , and z represent the three different bonds: x -bond, y -bond, and z -bond, respectively.

depending on the bond type, as shown in Fig. 22 (a). The first term represents the bond-dependent Kitaev interactions as discussed in Eq. (21). However, here we consider all three Kitaev interactions are equal to $K_x = K_y = K_z = K$. The second term, denoted by J , represents isotropic Heisenberg interaction.

For a ladder system, the Hamiltonian can be written as:

$$\begin{aligned} \mathcal{H} = & K \sum_{i=1}^{L/2} (S_{2i-1}^x S_{2i,1}^x + S_{2i,1}^y S_{2i+1,1}^y) + J \sum_{i=1}^L \vec{S}_{i,1} \cdot \vec{S}_{i+1,1} \\ & + K \sum_{i=1}^{L/2} (S_{2i,2}^x S_{2i+1,2}^x + S_{2i-1,2}^y S_{2i,2}^y) + J \sum_{i=1}^L \vec{S}_{i,2} \cdot \vec{S}_{i+1,2} \\ & + K \sum_{i=1}^L S_{i,1}^z S_{i,2}^z + J \sum_{i=1}^L \vec{S}_{i,1} \cdot \vec{S}_{i,2} \end{aligned} \quad (22)$$

The parameters J and K can be written in terms of ϕ as $J = \cos(\phi)$ and $K = \sin(\phi)$, where $\phi \in [0, 2\pi)$. The Hamiltonian can be rewritten in terms of raising and lowering operators as:

$$\begin{aligned} \mathcal{H}_{leg} = & J_L^T \sum_{j=1}^2 \sum_{i=1}^L (S_{i,j}^+ S_{i+1,j}^- + S_{i,j}^- S_{i+1,j}^+) \\ & + J_L^{TSF} \sum_{j=1}^2 \sum_{i=1}^L (-1)^{i+j} (S_{i,j}^+ S_{i+1,j}^+ + S_{i,j}^- S_{i+1,j}^-) \\ & + J_L^L \sum_{j=1}^2 \sum_{i=1}^L (S_{i,j}^z S_{i+1,j}^z) \end{aligned} \quad (23)$$

$$\mathcal{H}_{rung} = J_R^T \sum_{i=1}^L (S_{i,1}^+ S_{i,2}^- + S_{i,1}^- S_{i,2}^+) + J_R^L \sum_{i=1}^L S_{i,1}^z S_{i,2}^z \quad (24)$$

Here, $J_L^T = (2 \cos(\phi) + \sin(\phi))/4$, $J_L^{TSF} = \sin(\phi)/4$, $J_L^L = \cos(\phi)$, $J_R^T = \cos(\phi)/2$, and $J_R^L = (\cos(\phi) + \sin(\phi))$. The $SU(2)$ symmetry of the Hamiltonian is broken due to the anisotropy in the exchange interaction for any finite value of K . The system does not preserve the total spin S and its z component. Inversion symmetry is also not preserved for any finite value of K .

Quantum phases: The ground state phase diagram of the Kitaev-Heisenberg (KH) ladder exhibits a strong resemblance to that of the 2D honeycomb-lattice KH model. In the literature, the phase diagram of the KH model in the ϕ -dependent parameter space is characterized by several distinct phases: a broad rung singlet (RS) phase ($\phi/\pi = 0.0 - 0.48$ and $\phi/\pi = 1.74 - 2.0$), an antiferromagnetic Kitaev (AFK) phase ($\phi/\pi = 0.48 - 0.53$), a zigzag (ZZ) phase ($\phi/\pi = 0.53 - 0.81$), a ferromagnetic (FM) phase ($\phi/\pi = 0.81 - 1.36$), a ferromagnetic Kitaev (FK) phase ($\phi/\pi = 1.36 - 1.57$), and a stripe (ST) phase ($\phi/\pi = 1.56 - 1.72$) [443, 444]. Notable differences between the KH ladder and the 2D honeycomb-lattice KH model include the replacement of the Néel phase with the RS phase and an expanded range for the Kitaev spin liquid (KSL) phases (AFK and FK) in the ladder system. This expansion can be attributed to enhanced quantum fluctuations resulting from the reduced dimensionality of the ladder. The ground states of the RS, ZZ, FM, and ST phases exhibit long-range order, with the RS phase showing dimer-dimer correlations, and the ZZ, FM, and ST phases exhibiting longitudinal spin correlations. In contrast, the quantum spin liquid ground states (AFK and FK) show short-range spin correlations but instead exhibit long-range string order [444]. A detailed description of these phases is provided below.

Rung singlet phase: The ground state of an isotropic Heisenberg antiferromagnetic (HAF) spin- $\frac{1}{2}$ ladder system form a singlet rung dimer regardless of the strength of the rung interaction. The Hamiltonian of each rung in Eq. (24) conserves the total S^z . If each rung dimer is isolated, the singlet-triplet gap of each dimer is $\Delta = J = \cos\phi$ [443]. In other words, the ST gap is highest at $\phi = 0$, and it decreases as we move away from that point. In the full system, at $\phi = 0$, the KH model in Eq. (21) reduces to an isotropic HAF model with uniform exchange J which has the ST gap of $\Delta = 0.504J$ [445], which is smaller than the ST gap of an isolated dimer. The dimer phase in isotropic HAF coupled spin- $\frac{1}{2}$ ladder exhibits a finite energy

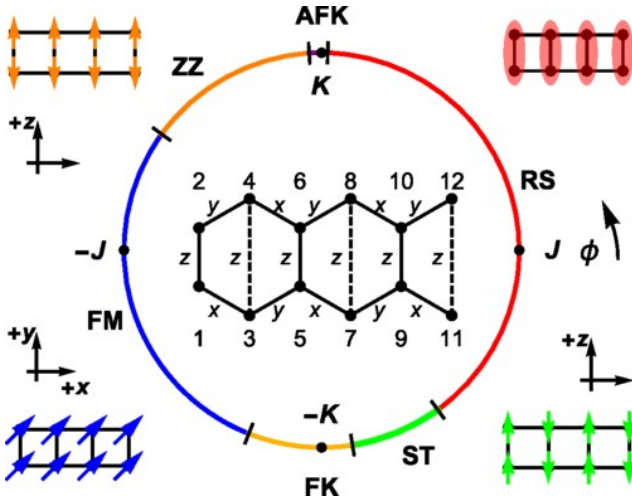


Figure 23. Phase diagram of the Kitaev-Heisenberg (KH) model on a two-leg ladder as a function of ϕ , with phase boundaries indicated by black bars. The diagram identifies six phases, including four long-range ordered phases: rung singlet (RS), zigzag (ZZ), stripe (ST), and ferromagnetic (FM), as well as two spin liquid phases: antiferromagnetic Kitaev (AFK) and ferromagnetic Kitaev (FK). Spin configurations are illustrated with colored spins. Inside the diagram, the two-leg ladder is shown, with bond definitions for the Kitaev term in Eq. (21), where bond-dependent interactions are labeled as x , y , and z . This figure is reproduced from Ref. [444].

gap and finite spin-spin correlations with exponential decay [445].

Stripy phase: In the region of $3/2\pi < \phi < 7/4\pi$, since $J > 0$ and $J + K < 0$, the leg and rung interactions are AFM and FM, respectively. This causes the rung spins to align ferromagnetically while the spins along the leg align antiferromagnetically. Transverse correlations in the gs of this system are extremely small, and longitudinal correlations manifest long-range Ising order. In this phase, the gs can be approximated as a product of Ising rung triplet states. For $\phi \approx 0.65\pi$, we could assume the wave function at $\phi \approx 1.65\pi$ to be

$$\psi(\phi) \approx \frac{1}{\sqrt{2}} \left[\prod_{i=1}^{N/2} S_{2i,1}^- S_{2i,2}^- |\uparrow\uparrow\rangle + S_{2i,1}^+ S_{2i,2}^+ \prod_{i=1}^{N/2} |\downarrow\downarrow\rangle \right]. \quad (25)$$

Here, $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ represent the spin-up and spin-down states, respectively. It is important to note that all spins are aligned along the z -axis. When the Hamiltonian given in Eq. (22) is applied to the wave function in Eq. (25), only the Ising interaction terms contribute nonzero components. As a result, no quantum fluctuations are present, and the system remains in a perfectly stripy state as described by Eq. (25).

Zigzag phase: In the region $\frac{1}{2\pi} < \phi < \frac{3}{4\pi}$, where $J < 0$ and $J + K > 0$, the interactions along the legs are ferromagnetic (FM), while the rung interactions

are antiferromagnetic (AFM). As a result, the spins on each leg are aligned ferromagnetically, but the spins on opposite legs are antiparallel. This phase is referred to as the “zigzag state,” following the terminology used for the corresponding phase in the honeycomb lattice Kitaev-Heisenberg (KH) model [446].

The ground state properties of this phase can be analyzed by examining its wave function. At $\phi \approx 0.65\pi$, the wave function can be expressed as:

$$\begin{aligned} \psi(\phi) \approx & [(C_1 + \sum_{m=1}^{N/2} C_2 (-1)^m (S_{m,1}^+ S_{m+1,1}^+ + S_{m,2}^- S_{m+1,2}^-) \\ & + C_3 S_{m,1}^+ S_{m,2}^-) + \sum_{m \neq n=1}^{N/2} C_4 S_{m,1}^+ S_{m+1,1}^+ S_{n,1}^+ S_{n+1,1}^+ \\ & + \dots] \prod_{i=1}^{N/2} S_{i,1}^- |\uparrow\rangle \\ & + [(C_1 + \sum_{m=1}^{N/2} C_2 (-1)^m (S_{m,1}^- S_{m+1,1}^- + S_{m,2}^+ S_{m+1,2}^+) \\ & + C_3 S_{m,1}^- S_{m,2}^+) + \sum_{m \neq n=1}^{N/2} C_4 S_{m,1}^- S_{m+1,1}^- S_{n,1}^- S_{n+1,1}^- \\ & + \dots] \prod_{i=1}^{N/2} S_{i,1}^+ |\downarrow\rangle + \dots \end{aligned} \quad (26)$$

In this expression, $|\uparrow\rangle$ and $|\downarrow\rangle$ denote the spin-up and spin-down states, respectively. The dominant term, with coefficient C_1 , corresponds to two configurations where the spins on one leg are aligned along the $+z$ -direction, while the spins on the other leg are aligned along the $-z$ -direction. The next most significant terms, with coefficients C_2 and C_3 , describe configurations involving a double spin flip along the leg and rung, with C_2 contributing to $2N$ configurations and C_3 contributing to N configurations. The coefficient C_4 corresponds to $2N$ configurations involving a simultaneous four-spin flip along the leg. The contributions from other configurations, which involve multiple spin flips, are comparatively smaller.

FM phase: The FM phase occurs in the range $0.81\pi < \phi < 1.37\pi$, and at $\phi = \pi$ the FM state is isotropic with $SU(2)$ symmetry which gives rise to $(N+1)$ fold degeneracy. The longitudinal correlation is close to 0 in the $S^z = 0$ sector and close to $1/4$ in the $S^z = N/2$ sector. On the other hand, the transverse correlation is close to $1/4$ in the $S^z = 0$ sector and close to 0 in the $S^z = N/2$ sector. For any deviation from $\phi = \pi$, the finite value of the two-spin flip exchange lifts the $(N+1)$ -fold degeneracy of the isotropic system, and the gs becomes anisotropic. The XY correlations dominate in this phase; therefore, the FM phase is called the XY-FM phase.

Kitaev spin liquid: The long-range ordered phases occupy the majority of the ϕ range. In the remaining regions around the Kitaev points $K = \pm\pi/2$, no long-range ordering is observed, suggesting the presence of spin-liquid states. To explore the similarities between these states and the so-called Kitaev spin liquid (KSL) in the honeycomb Kitaev-Heisenberg model [427], one can compute the expectation value of the plaquette operator.

Plaquette operator: The gs of the Kitaev model in a hexagonal cluster or ladder is the π -flux state. The expectation value of the plaquette operator $\mathcal{O}_{\text{plaquette}}$ in this phase has a value of 1. We define the plaquette operator for the system, as described in Eq. (21), as:

$$\mathcal{O}_{\text{plaquette}} = \langle S_1^x S_2^y S_3^z S_4^x S_5^y S_6^z \rangle \quad (27)$$

Fig.6 (a) and (b) of Ref. [443] show the $\mathcal{O}_{\text{plaquette}}$ calculated in ED. At both the Kitaev points $\phi = \pm\pi/2$, $\mathcal{O}_{\text{plaquette}}$ is 1 as expected. As we move away from $\phi = \pm\pi/2$, $\mathcal{O}_{\text{plaquette}}$ decreases from 1 and rapidly decays to zero near the long-range ordered phases.

String order parameter: To investigate the topological properties of a spin system, one can examine hidden order. In particular, as discussed in previous studies [444, 447, 448], string order is anticipated in Kitaev systems. To numerically confirm this, the string order parameter can be calculated, which is defined as:

$$\mathcal{O}_{\text{string}} = \lim_{|i-j| \rightarrow \infty} \left\langle S_{i,1}^y S_{i,2}^x \left(\prod_{k=i+1}^{j-1} S_{k,1}^z S_{k,2}^z \right) S_{j,1}^y S_{j,2}^x \right\rangle \quad (28)$$

The extrapolated string order parameters near the ferromagnetic (FM) Kitaev point are shown as a function of $\frac{\phi}{\pi}$ in Fig. 8 of Ref. [443]. The presence of long-range string order is clearly observed throughout the entire FM spin-liquid (SL) phase, with the maximum occurring at the FM Kitaev point.

Quadrupolar order in KH ladder: The ground-state phase diagram of the honeycomb lattice Kitaev-Heisenberg (KH) model has been extensively investigated. However, the potential for quantum phases involving higher spin states, such as quadrupolar (QP) or nematic order, within the KH model remains largely unexplored. This raises the question of whether QP order can emerge in a pure spin- $\frac{1}{2}$ Kitaev-Heisenberg model on both ladder and honeycomb lattice geometries.

In this study [449], we investigate the properties of the quadrupolar phase in the KH model on two distinct geometries: the two-legged ladder and the two-dimensional honeycomb lattice. We employ exact diagonalization (ED) and density matrix

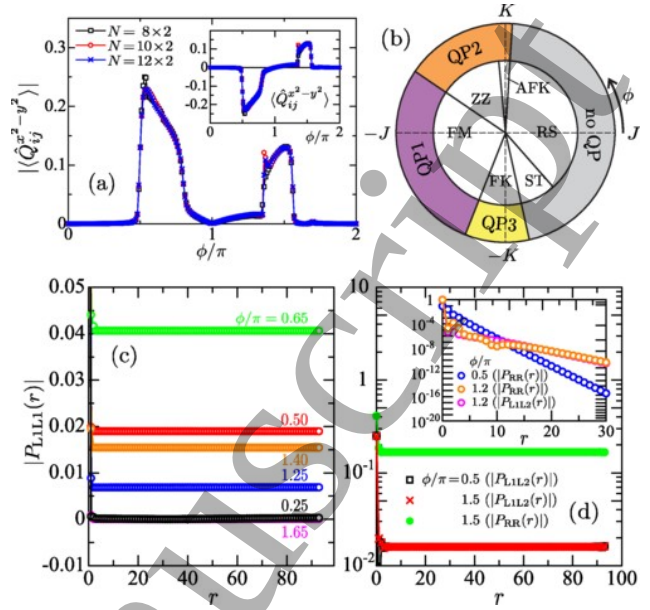


Figure 24. Quadrupolar results for the two-leg Kitaev-Heisenberg (KH) ladder. (a) Absolute values of the quadrupolar (QP) order parameter $\hat{Q}_{ij}^{x^2-y^2} = \frac{1}{2}(S_{i,j}^+ S_{i+1,j}^+ + S_{i,j}^- S_{i+1,j}^-)$ for leg bonds across different system sizes. The inset shows the same physical quantity without the absolute values. (b) Phase diagram depicting magnetic order (inner) and QP order (outer). The diagram includes four long-range ordered phases: rung-singlet (RS), zigzag (ZZ), ferromagnetic (FM), and stripy (ST), as well as two quantum spin liquid (QSL) phases: antiferromagnetic Kitaev (AFK) and ferromagnetic Kitaev (FK). (c) QP correlation functions $|P_{L1L1}(r)| = |\langle S_{i,1}^+ S_{i+1,1}^+ S_{i+r,1}^- S_{i+1+r,1}^- \rangle|$ for a $N = 100 \times 2$ cluster with representative parameters. (d) Interleg and rung-rung QP correlation functions at $\phi/\pi = 0.5, 1.2$, and 1.5 . The inset provides examples where the QP correlation function exhibits an exponential decay. This figure is reproduced from Ref. [449].

renormalization group (DMRG) techniques to analyze the ground-state properties. Our results present a comprehensive phase diagram, as shown in Fig. 24, demonstrating the stability of quadrupolar order over a broad region of the phase diagram, particularly near the Kitaev spin-liquid (KSL) phases. This stability is especially notable as it occurs in the absence of long-range spin-spin correlations, suggesting a complex relationship between QP order and QSLs. Notably, a positive third-order susceptibility, which serves as a general signature of QP order, has been observed near the FK QSL phase in α -RuCl₃ [450, 451]. These findings not only enhance our understanding of the KH model but also contribute to the broader field of quantum magnetism in low-dimensional systems, shedding light on the intricate interplay between various types of quantum order.

10. Future perspective

In this topical review, we explore the intriguing world of low-dimensional spin- $\frac{1}{2}$ systems, focusing on a selected range of models that have garnered significant attention in both theoretical and experimental research. Specifically, we delve into the frustrated one-dimensional spin- $\frac{1}{2}$ MG model, as well as the Heisenberg spin- $\frac{1}{2}$ model applied to diverse ladder geometries, including normal, zigzag, honeycomb, skewed, and diamond ladder. Our discussion not only provided references to the theoretical studies of these models but also investigates their experimental realizations in real materials, providing a comprehensive view of how these theoretical concepts are being tested in the laboratories. We also present tables summarizing the studied materials and their corresponding model parameters.

In some cases, such as the Heisenberg model on skewed ladder and the Heisenberg-Kitaev model on the ladder system, we focus on the theoretical studies to inspire experimental scientists to develop and test new materials that could be the possible realization of model. The spin-liquid phases in both the antiferromagnetic and ferromagnetic Kitaev phases on the ladder show remarkable similarities to those observed on the honeycomb lattice, offering promising opportunities for new insights into quantum spin liquids. Additionally, the plateau phases and exotic quantum states in the ground state of the isotropic Heisenberg model on the skewed ladder remain largely unexplored, presenting an exciting open frontier in the field.

The spin- $\frac{1}{2}$ MG model chain is another system brimming with potential, as it encompasses a wide and largely uncharted parameter space, making the design of suitable materials both a challenging and rewarding task. This vast landscape of possibilities calls for further experimental innovation to fully unlock the potential of these systems.

Another promising approach for investigating quantum phases is the use of quantum simulators [452–455]. These platforms enable the mapping of spin- $\frac{1}{2}$ degrees of freedom onto qubits, allowing the simulation of Hamiltonians using various quantum algorithms such as the Quantum Approximate Optimization Algorithm (QAOA) [456–463], the Variational Quantum Eigensolver (VQE) [464–470], and the Sampling Quantum Diagonalization (SQD) [471–473]. To characterize quantum phase transitions, several diagnostic tools such as entanglement entropy [474–480], fidelity susceptibility [481–484], and concurrence [485–487] serve as insightful measures. However, a detailed discussion of these diagnostic tools lies beyond the scope of this review. Interested readers are referred to Refs. [488–493] for further information.

In conclusion, this review highlights selected

isotropic Heisenberg spin- $\frac{1}{2}$ model systems on chains and ladders, offering an in-depth look at both theoretical predictions and their experimental realizations. We also shed light on the fascinating quantum phases—such as spin liquids, plateau phases, and other exotic ground states, that are yet to be realized in real materials. These unexplored phenomena provide an exciting playground for future research, urging the scientific community to push the boundaries of what is possible in the realm of low-dimensional quantum systems.

Acknowledgment

M.K. thanks his collaborators professor Z. G. Soos, professor S. Ramasesha, professor D. Sen, professor T. Dasgupta, Professor M. K. Harbola, professor J. van den Brink, professor S. Nishimoto, professor H. Katsura, professor R. R. P Singh, professor A. Mukherjee, professor S. M. Yusuf, professor A. K. Bera, professor R. Nath, Dr. S. K. Saha, Dr. A. Parvej, miss M. Chatterjee, Dr. D. Maiti, Dr. D. Dey, Dr. S. Das, and Dr. G. Giri for their contribution in various projects on spin- $\frac{1}{2}$ systems. MK also acknowledges support from SERB (ANRF) India through Grant Sanction No. CRG/2020/000754.

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